Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Five H Farms

Physical address of dairy:

Number and Street	City	County	Zip Code
5851 S Highway 59	Merced	Merced	95340

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/01/1999

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X065-X130-X051-XXXX

B. OPERATORS

Vyfhuisen, Case

Operator name: Vyfhuisen, Case

Telephone no.: (209) 631-1887 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5851 S Hwy 59	Merced	CA	95340

This operator is responsible for paying permit fees.

C. OWNERS

Vyfhuisen, Case

Legal owner name: Vyfhuisen, Case

Telephone no.: (209) 631-1887 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5851 S Hwy 59	Merced	CA	95340

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	580	0	300	0
Number under roof	2,060	340	0	0	0	0
Maximum number	2,146	350	700	0	300	0
Average number	2,060	340	580	0	300	0
Avg live weight (lbs)	1,400	1,400	900	0		

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 66,600.12 *tons per reporting period*

Total nitrogen from manure: 848,592.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 594,015.04 *lbs per reporting period*

Total phosphorus from manure: 140,201.54 *lbs per reporting period*

Total potassium from manure: 405,336.37 *lbs per reporting period*

Total salt from manure: 1,048,134.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 46,591,000 *gallons*

Total nitrogen generated: 311,581.37 *lbs*

Total phosphorus generated: 53,842.36 *lbs*

Total potassium generated: 314,263.13 *lbs*

Total salt generated: 1,534,561.30 *lbs*

	46,591,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	46,591,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Wells	Ground water
IW 1	Ground water
IW 10	Ground water
IW 11	Ground water
IW 12	Ground water

Source Description	Type
IW 2	Ground water
IW 3	Ground water
IW 4	Ground water
IW 5	Ground water
IW 6	Ground water
IW 7	Ground water
IW 8	Ground water
IW 9	Ground water
IW Borba	Ground water
MID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/26/2023	Solid commercial fertilizer *UN 32*	27.60 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	17,664.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	17,664.00	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

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APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	35	35	2	process wastewater	X065-X130-X051-XXXX
Field 10-11	151	151	2	manure	X074-X080-X023-XXXX
Field 12	75	75	1	none	X074-X080-X020-XXXX
Field 15	111	111	2	process wastewater	X066-X160-X003-XXXX
Field 16	71	71	2	both	X066-X160-X003-XXXX
Field 17	84	84	1	process wastewater	X066-X160-X005-XXXX
Field 18	75	75	1	process wastewater	X066-X160-X017-XXXX
Field 19	54	54	2	manure	X066-X160-X017-XXXX
Field 2	52	52	2	process wastewater	X065-X130-X051-XXXX
Field 20	29	29	2	both	X066-X160-X010-XXXX X066-X160-X017-XXXX
Field 21	25	25	2	both	X066-X160-X010-XXXX X066-X160-X017-XXXX X066-X160-X018-XXXX
Field 29	34	34	2	process wastewater	X065-X130-X044-XXXX
Field 3	38	38	2	process wastewater	X065-X130-X051-XXXX
Field 30	42	42	2	process wastewater	X065-X130-X044-XXXX
Field 31	36	36	2	both	X065-X130-X044-XXXX
Field 32	41	41	2	both	X065-X130-X044-XXXX
Field 4	35	35	2	process wastewater	X065-X130-X051-XXXX
Field 5	33	33	2	process wastewater	X065-X130-X051-XXXX
Field 6	50	50	1	process wastewater	X065-X130-X051-XXXX
Field 7	44	44	2	process wastewater	X065-X130-X051-XXXX
Field 8	22	22	0	none	X074-X080-X020-XXXX
Totals for areas that were used for application	1,040	1,040	35		
Totals for areas that were not used for application	97	97	1		
Land application area totals	1,137	1,137	36		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

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Field 1

10/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 35 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	632.70 *ton*	As-is		55.4	7,500.00	1,300.00	11,000.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	18.08	271.16	47.00	397.70	1,354.48

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/05/2023	808.00 *ton*	As-is		63.9	4,500.00	900.00	5,600.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	23.09	207.77	41.55	258.56	933.40

Field 10-11

Field name: Field 10-11

10/26/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 151 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	683.80 *ton*	As-is		11.0	9,400.00	2,000.00	9,900.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	4.53	85.14	18.11	89.66	1,088.19

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Field 10-11

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 151 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	3,387.60 *ton*	As-is		57.7	4,300.00	900.00	3,500.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	22.43	192.94	40.38	157.04	873.06

Field 12

Field name: Field 12

10/01/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 75 Plant date: 10/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	771.98 *ton*	As-is		10.0	24,200.00	2,900.00	15,300.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	10.29	498.18	59.70	314.97	1,556.31

Field 15

Field name: Field 15

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Field 15

10/20/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 111 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	1,245.00 *ton*	As-is		64.3	5,100.00	900.00	4,500.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	11.22	114.41	20.19	100.95	848.89

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 111 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	2,857.50 *ton*	As-is		63.3	5,200.00	900.00	3,200.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	25.74	267.73	46.34	164.76	1,133.73

Field 16

Field name: Field 16

10/20/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	1,334.50 *ton*	As-is		63.9	5,100.00	1,000.00	5,300.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	19.06	194.46	38.13	202.08	1,321.38

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Field 16

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	1,712.40 *ton*	As-is		63.6	3,900.00	900.00	2,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	24.46	190.81	44.03	141.88	1,032.92

Field 17

Field name: Field 17

09/25/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 84 Plant date: 09/25/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	950.40 *ton*	As-is		10.0	24,300.00	2,900.00	15,300.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	11.31	549.87	65.62	346.22	1,934.74

Field 18

Field name: Field 18

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Field 18

10/12/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 75 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	700.30 *ton*	As-is		10.0	32,900.00	3,300.00	23,700.00		16.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.34	614.40	61.63	442.59	2,697.56

Field 19

Field name: Field 19

10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 54 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	923.20 *ton*	As-is		51.3	5,400.00	1,500.00	8,100.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	17.10	184.64	51.29	276.96	1,415.40

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	1,258.90 *ton*	As-is		66.3	3,700.00	800.00	2,800.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	23.31	172.52	37.30	130.55	738.51

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Field 2

Field name: Field 2

10/15/2022: Rye Silage

Crop: Rye Silage Acres planted: 52 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	654.90 *ton*	As-is		61.7	10,800.00	1,800.00	14,900.00		13.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	21.00	210.00	25.20	252.00	0.00
Total actual harvest content	12.59	272.04	45.34	375.31	1,316.84

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	832.90 *ton*	As-is		61.1	5,100.00	1,000.00	8,800.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	16.02	163.38	32.03	281.90	984.46

Field 20

Field name: Field 20

10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 29 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	369.30 *ton*	As-is		61.3	4,700.00	1,100.00	5,600.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	12.73	119.70	28.02	142.63	1,044.79

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Field 20

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	688.80 *ton*	As-is		63.9	4,000.00	700.00	2,900.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	23.75	190.01	33.25	137.76	840.29

Field 21

Field name: Field 21

10/21/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 25 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	317.10 *ton*	As-is		58.6	4,300.00	1,200.00	5,100.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	12.68	109.08	30.44	129.38	903.20

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	604.90 *ton*	As-is		63.8	4,200.00	900.00	3,100.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	24.20	203.25	43.55	150.02	945.97

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Field 29

Field name: Field 29

10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 34 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	408.40 *ton*	As-is		60.9	4,400.00	1,000.00	5,900.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	12.01	105.70	24.02	141.74	967.50

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	480.20 *ton*	As-is		60.5	4,500.00	1,000.00	3,500.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	14.12	127.11	28.25	98.86	513.25

Field 3

Field name: Field 3

09/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 38 Plant date: 09/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	600.30 *ton*	As-is		54.9	6,900.00	1,300.00	10,300.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	15.80	218.00	41.07	325.43	1,396.42

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Field 3

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/05/2023	943.30 *ton*	As-is		71.7	4,100.00	600.00	4,800.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	24.82	203.55	29.79	238.31	941.36

Field 30

Field name: Field 30

10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 42 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	500.10 *ton*	As-is		59.1	4,300.00	1,000.00	6,100.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	11.91	102.40	23.81	145.27	1,022.70

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	665.40 *ton*	As-is		61.5	4,100.00	1,000.00	3,700.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	22.18	181.88	44.36	164.13	853.93

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Field 31

Field name: Field 31

10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 36 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	364.00 *ton*	As-is		56.9	8,800.00	1,600.00	11,300.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	10.11	177.96	32.36	228.51	793.14

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	584.90 *ton*	As-is		57.8	5,900.00	1,300.00	7,700.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	23.40	276.07	60.83	360.30	1,441.47

Field 32

Field name: Field 32

10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 41 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	636.30 *ton*	As-is		66.1	8,900.00	1,600.00	11,600.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	15.52	276.25	49.66	360.05	1,283.71

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Field 32

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	584.90 *ton*	As-is		62.9	5,500.00	1,200.00	5,600.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	23.40	257.36	56.15	262.04	1,284.63

Field 4

Field name: Field 4

09/11/2022: Rye Silage

Crop: Rye Silage Acres planted: 35 Plant date: 09/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	463.70 *ton*	As-is		70.6	9,500.00	1,600.00	12,900.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	21.00	210.00	25.20	252.00	0.00
Total actual harvest content	13.25	251.72	42.40	341.81	1,020.51

04/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 04/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	788.00 *ton*	As-is		69.4	4,300.00	800.00	5,700.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	22.51	193.62	36.02	256.66	909.40

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Field 5

Field name: Field 5

09/11/2022: Rye Silage

Crop: Rye Silage Acres planted: 33 Plant date: 09/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	389.70 *ton*	As-is		66.2	9,200.00	1,600.00	14,200.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	21.00	210.00	25.20	252.00	0.00
Total actual harvest content	11.81	217.29	37.79	335.38	997.87

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	943.40 *ton*	As-is		68.2	4,500.00	800.00	6,200.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	28.59	257.29	45.74	354.49	1,218.19

Field 6

Field name: Field 6

11/01/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 50 Plant date: 11/01/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	489.38 *ton*	As-is		10.0	30,300.00	3,300.00	30,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.79	593.13	64.60	604.87	1,920.33

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Field 7

Field name: Field 7

10/15/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 42 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	855.20 *ton*	As-is		59.4	5,800.00	1,200.00	8,800.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.50	159.50	24.65	120.35	0.00
Total actual harvest content	20.36	236.20	48.87	358.37	1,669.92

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,073.70 *ton*	As-is		63.1	3,600.00	13,400.00	3,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	200.00	0.00
Total actual harvest content	25.56	184.06	685.12	194.29	1,301.78

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NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/15/2022: Wheat, silage, soft dough

Field name: Field 1

Crop: Wheat, silage, soft dough — Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.97	6.67	112.11	630.52	645,000.00 *gal*
Irrigation Wells	Ground water	0.71	0.00	0.00	141.48	2,967,000.00 *gal*
Irrigation Wells	Ground water	13.68	0.00	0.00	492.12	2,064,000.00 *gal*
Application event totals		84.36	6.67	112.11	1,264.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	154.50	17.95	164.34	927.01	960,000.00 *gal*
Irrigation Wells	Ground water	0.81	0.00	0.00	157.29	3,383,040.00 *gal*
Application event totals		155.31	17.95	164.34	1,084.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.99	0.00	0.00	197.42	4,140,000.00 *gal*
Application event totals		0.99	0.00	0.00	197.42	

Field 1 - 05/26/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.60	5.50	99.86	533.75	615,000.00 *gal*
Irrigation Wells	Ground water	0.76	0.00	0.00	151.35	3,174,000.00 *gal*
Application event totals		68.35	5.50	99.86	685.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.82	0.00	0.00	164.52	3,450,000.00 *gal*
Application event totals		0.82	0.00	0.00	164.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.13	6.88	148.04	808.99	780,000.00 *gal*
Irrigation Wells	Ground water	0.92	0.00	0.00	184.26	3,864,000.00 *gal*
Irrigation Wells	Ground water	0.40	0.00	0.00	142.20	1,680,000.00 *gal*
Application event totals		84.45	6.88	148.04	1,135.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.76	0.00	0.00	151.35	3,174,000.00 *gal*
Application event totals		0.76	0.00	0.00	151.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.20	0.00	0.00	240.19	5,037,000.00 *gal*
Application event totals		1.20	0.00	0.00	240.19	

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Field 1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.70	10.32	222.05	1,213.48	1,170,000.00 *gal*
Irrigation Wells	Ground water	1.58	0.00	0.00	315.87	6,624,000.00 *gal*
Application event totals		126.28	10.32	222.05	1,529.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.46	0.00	0.00	292.84	6,141,000.00 *gal*
Irrigation Wells	Ground water	0.64	0.00	0.00	225.99	2,670,000.00 *gal*
Application event totals		2.10	0.00	0.00	518.83	

Field 10-11 - 10/26/2022: Wheat Hay

Field name: Field 10-11

Crop: Wheat Hay

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	94.16	11.99	33.59	0.00	1,600.00 *ton*
Application event totals		94.16	11.99	33.59	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.51	0.00	0.00	169.96	9,180,000.00 *gal*
Irrigation Wells	Ground water	0.68	0.00	0.00	233.37	12,240,000.00 *gal*
Application event totals		1.18	0.00	0.00	403.33	

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Field 10-11 - 10/26/2022: Wheat Hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.63	0.00	0.00	209.95	11,340,000.00 *gal*
Irrigation Wells	Ground water	0.84	0.00	0.00	288.28	15,120,000.00 *gal*
Application event totals		1.46	0.00	0.00	498.23	

Field 10-11 - 06/18/2023: Corn, silage

Field name: Field 10-11

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	191.58	29.69	192.53	0.00	3,200.00 *ton*
Application event totals		191.58	29.69	192.53	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.72	0.00	0.00	239.94	12,960,000.00 *gal*
Irrigation Wells	Ground water	0.95	0.00	0.00	329.47	17,280,000.00 *gal*
Application event totals		1.67	0.00	0.00	569.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.91	0.00	0.00	303.92	16,416,000.00 *gal*
Irrigation Wells	Ground water	1.21	0.00	0.00	417.33	21,888,000.00 *gal*
Application event totals		2.12	0.00	0.00	721.25	

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Field 10-11 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.45	0.00	0.00	151.96	8,208,000.00 *gal*
Irrigation Wells	Ground water	0.60	0.00	0.00	208.66	10,944,000.00 *gal*
Application event totals		1.06	0.00	0.00	360.62	

08/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.54	0.00	0.00	179.95	9,720,000.00 *gal*
Irrigation Wells	Ground water	0.72	0.00	0.00	247.10	12,960,000.00 *gal*
Application event totals		1.25	0.00	0.00	427.05	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.45	0.00	0.00	149.96	8,100,000.00 *gal*
Irrigation Wells	Ground water	0.60	0.00	0.00	205.92	10,800,000.00 *gal*
Application event totals		1.04	0.00	0.00	355.88	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.83	0.00	0.00	285.54	14,976,000.00 *gal*
Application event totals		0.83	0.00	0.00	285.54	

Field 12 - 10/01/2020: Alfalfa, hay

Field name: Field 12

Crop: Alfalfa, hay

Plant date: 10/01/2020

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Field 12 - 10/01/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	9.45	0.00	0.00	3,261.36	84,960,000.00 *gal*
Application event totals		9.45	0.00	0.00	3,261.36	

Field 15 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 15

Crop: Wheat, silage, soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	130.36	36.23	56.36	117.33	510,000.00 *gal*
Application event totals		130.36	36.23	56.36	117.33	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.62	0.00	0.00	113.29	8,280,000.00 *gal*
Irrigation Wells	Ground water	0.31	0.00	0.00	57.58	4,140,000.00 *gal*
Application event totals		0.93	0.00	0.00	170.87	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.67	0.00	0.00	121.50	8,880,000.00 *gal*
Application event totals		0.67	0.00	0.00	121.50	

Field 15 - 05/31/2023: Corn, silage

Field name: Field 15

Crop: Corn, silage

Plant date: 05/31/2023

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Field 15 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	325.91	90.58	140.91	293.32	1,275,000.00 *gal*
Application event totals		325.91	90.58	140.91	293.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.41	0.00	0.00	75.11	5,400,000.00 *gal*
MID Canal	Surface water	0.81	0.00	0.00	201.36	10,800,000.00 *gal*
Application event totals		1.22	0.00	0.00	276.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.37	0.00	0.00	68.01	4,890,000.00 *gal*
MID Canal	Surface water	0.74	0.00	0.00	182.35	9,780,000.00 *gal*
Application event totals		1.10	0.00	0.00	250.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.39	0.00	0.00	72.18	5,190,000.00 *gal*
MID Canal	Surface water	0.78	0.00	0.00	193.53	10,380,000.00 *gal*
Application event totals		1.17	0.00	0.00	265.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

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Field 15 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.34	0.00	0.00	62.59	4,500,000.00 *gal*
MID Canal	Surface water	0.68	0.00	0.00	167.80	9,000,000.00 *gal*
Application event totals		1.01	0.00	0.00	230.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.36	0.00	0.00	66.34	4,770,000.00 *gal*
MID Canal	Surface water	0.72	0.00	0.00	177.87	9,540,000.00 *gal*
Application event totals		1.08	0.00	0.00	244.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.37	0.00	0.00	68.85	4,950,000.00 *gal*
MID Canal	Surface water	0.74	0.00	0.00	184.58	9,900,000.00 *gal*
Application event totals		1.12	0.00	0.00	253.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.38	0.00	0.00	70.52	5,070,000.00 *gal*
MID Canal	Surface water	0.76	0.00	0.00	189.06	10,140,000.00 *gal*
Application event totals		1.14	0.00	0.00	259.57	

Field 16 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 16

Crop: Wheat, silage, soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 16 - 10/20/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	258.40	71.82	111.72	232.56	637,500.00 *gal*
Application event totals		258.40	71.82	111.72	232.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.74	0.00	0.00	135.39	6,240,000.00 *gal*
Irrigation Wells	Ground water	0.37	0.00	0.00	68.81	3,120,000.00 *gal*
Application event totals		1.12	0.00	0.00	204.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.64	0.00	0.00	117.16	5,400,000.00 *gal*
Irrigation Wells	Ground water	0.32	0.00	0.00	59.55	2,700,000.00 *gal*
Application event totals		0.97	0.00	0.00	176.71	

Field 16 - 06/20/2023: Corn, silage

Field name: Field 16

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	219.54	34.03	220.64	0.00	1,700.00 *ton*
Application event totals		219.54	34.03	220.64	0.00	

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Field 16 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.06	0.00	0.00	192.67	8,880,000.00 *gal*
Irrigation Wells	Ground water	0.53	0.00	0.00	97.92	4,440,000.00 *gal*
Application event totals		1.59	0.00	0.00	290.59	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.03	0.00	0.00	187.46	8,640,000.00 *gal*
Irrigation Wells	Ground water	0.52	0.00	0.00	95.28	4,320,000.00 *gal*
Application event totals		1.55	0.00	0.00	282.74	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.39	0.00	0.00	71.46	3,240,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	191.58	6,480,000.00 *gal*
Application event totals		1.16	0.00	0.00	263.04	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.34	0.00	0.00	63.52	2,880,000.00 *gal*
MID Canal	Surface water	0.69	0.00	0.00	170.30	5,760,000.00 *gal*
Application event totals		1.03	0.00	0.00	233.81	

08/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

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Field 16 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.21	0.00	0.00	38.38	1,740,000.00 *gal*
MID Canal	Surface water	0.41	0.00	0.00	102.89	3,480,000.00 *gal*
Application event totals		0.62	0.00	0.00	141.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.17	0.00	0.00	31.76	1,440,000.00 *gal*
MID Canal	Surface water	0.34	0.00	0.00	85.15	2,880,000.00 *gal*
Application event totals		0.52	0.00	0.00	116.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.34	0.00	0.00	63.69	2,888,000.00 *gal*
MID Canal	Surface water	0.69	0.00	0.00	170.30	5,760,000.00 *gal*
Application event totals		1.03	0.00	0.00	233.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.26	0.00	0.00	47.64	2,160,000.00 *gal*
MID Canal	Surface water	0.52	0.00	0.00	127.72	4,320,000.00 *gal*
Application event totals		0.77	0.00	0.00	175.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.19	0.00	0.00	35.73	1,620,000.00 *gal*
MID Canal	Surface water	0.39	0.00	0.00	95.79	3,240,000.00 *gal*
Application event totals		0.58	0.00	0.00	131.52	

Field 17 - 09/25/2018: Alfalfa, hay

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Field 17 - 09/25/2018: Alfalfa, hay

Field name: Field 17

Crop: Alfalfa, hay

Plant date: 09/25/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	321.56	89.37	139.03	289.40	952,000.00 *gal*
Application event totals		321.56	89.37	139.03	289.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.28	0.00	0.00	51.28	2,790,000.00 *gal*
Irrigation Wells	Ground water	0.55	0.00	0.00	134.71	5,580,000.00 *gal*
Application event totals		0.83	0.00	0.00	185.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.38	0.00	0.00	70.57	3,840,000.00 *gal*
Irrigation Wells	Ground water	0.76	0.00	0.00	185.40	7,680,000.00 *gal*
Application event totals		1.14	0.00	0.00	255.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.36	0.00	0.00	66.16	3,600,000.00 *gal*
Irrigation Wells	Ground water	0.72	0.00	0.00	173.81	7,200,000.00 *gal*
Application event totals		1.07	0.00	0.00	239.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.57	0.00	0.00	104.15	5,760,000.00 *gal*
Irrigation Wells	Ground water	0.29	0.00	0.00	52.93	2,880,000.00 *gal*
Application event totals		0.86	0.00	0.00	157.08	

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Field 17 - 09/25/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.70	0.00	0.00	126.93	7,020,000.00 *gal*
Irrigation Wells	Ground water	0.35	0.00	0.00	64.51	3,510,000.00 *gal*
Application event totals		1.05	0.00	0.00	191.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.32	0.00	0.00	58.44	3,180,000.00 *gal*
MID Canal	Surface water	0.63	0.00	0.00	156.70	6,360,000.00 *gal*
Application event totals		0.95	0.00	0.00	215.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.43	0.00	0.00	78.11	4,320,000.00 *gal*
Irrigation Wells	Ground water	0.21	0.00	0.00	39.70	2,160,000.00 *gal*
Application event totals		0.64	0.00	0.00	117.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.37	0.00	0.00	68.37	3,720,000.00 *gal*
MID Canal	Surface water	0.74	0.00	0.00	183.30	7,440,000.00 *gal*
Application event totals		1.11	0.00	0.00	251.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.14	0.00	0.00	208.29	11,520,000.00 *gal*
Irrigation Wells	Ground water	0.57	0.00	0.00	105.86	5,760,000.00 *gal*
Application event totals		1.72	0.00	0.00	314.15	

Field 18 - 10/12/2022: Alfalfa, hay

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Field 18 - 10/12/2022: Alfalfa, hay

Field name: Field 18

Crop: Alfalfa, hay

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	373.01	103.67	161.27	335.71	986,000.00 *gal*
Application event totals		373.01	103.67	161.27	335.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.55	0.00	0.00	99.63	4,920,000.00 *gal*
Irrigation Wells	Ground water	0.27	0.00	0.00	50.64	2,460,000.00 *gal*
Application event totals		0.82	0.00	0.00	150.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.64	0.00	0.00	116.64	5,760,000.00 *gal*
Irrigation Wells	Ground water	0.32	0.00	0.00	59.28	2,880,000.00 *gal*
Application event totals		0.96	0.00	0.00	175.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.72	0.00	0.00	131.22	6,480,000.00 *gal*
Irrigation Wells	Ground water	0.36	0.00	0.00	66.69	3,240,000.00 *gal*
Application event totals		1.08	0.00	0.00	197.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.46	0.00	0.00	112.91	4,176,000.00 *gal*
Irrigation Wells	Ground water	0.29	0.00	0.00	53.73	2,610,000.00 *gal*
Application event totals		0.76	0.00	0.00	166.63	

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Field 18 - 10/12/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.61	0.00	0.00	147.95	5,472,000.00 *gal*
Irrigation Wells	Ground water	0.68	0.00	0.00	32.88	6,156,000.00 *gal*
Application event totals		1.29	0.00	0.00	180.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.61	0.00	0.00	149.25	5,520,000.00 *gal*
Irrigation Wells	Ground water	0.69	0.00	0.00	33.17	6,210,000.00 *gal*
Application event totals		1.31	0.00	0.00	182.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.71	0.00	0.00	172.61	6,384,000.00 *gal*
Irrigation Wells	Ground water	0.80	0.00	0.00	38.36	7,182,000.00 *gal*
Application event totals		1.51	0.00	0.00	210.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.74	0.00	0.00	180.40	6,672,000.00 *gal*
Irrigation Wells	Ground water	0.84	0.00	0.00	40.09	7,506,000.00 *gal*
Application event totals		1.58	0.00	0.00	220.48	

Field 19 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 19

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 19 - 10/21/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	197.47	25.16	70.44	0.00	1,200.00 *ton*
Application event totals		197.47	25.16	70.44	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.67	0.00	0.00	121.50	4,320,000.00 *gal*
Irrigation Wells	Ground water	0.33	0.00	0.00	61.75	2,160,000.00 *gal*
Application event totals		1.00	0.00	0.00	183.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.46	0.00	0.00	84.38	3,000,000.00 *gal*
Irrigation Wells	Ground water	0.23	0.00	0.00	42.88	1,500,000.00 *gal*
Irrigation Wells	Ground water	0.46	0.00	0.00	112.66	3,000,000.00 *gal*
Application event totals		1.16	0.00	0.00	239.92	

Field 19 - 06/03/2023: Corn, silage

Field name: Field 19

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	192.52	29.84	193.48	0.00	1,150.00 *ton*
Application event totals		192.52	29.84	193.48	0.00	

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Field 19 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.82	0.00	0.00	198.28	5,280,000.00 *gal*
Irrigation Wells	Ground water	0.92	0.00	0.00	44.06	5,940,000.00 *gal*
Application event totals		1.73	0.00	0.00	242.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.69	0.00	0.00	166.73	4,440,000.00 *gal*
Irrigation Wells	Ground water	0.55	0.00	0.00	133.39	3,552,000.00 *gal*
Irrigation Wells	Ground water	0.62	0.00	0.00	29.64	3,996,000.00 *gal*
Application event totals		1.85	0.00	0.00	329.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.63	0.00	0.00	153.21	4,080,000.00 *gal*
Irrigation Wells	Ground water	0.50	0.00	0.00	122.57	3,264,000.00 *gal*
Irrigation Wells	Ground water	0.57	0.00	0.00	27.24	3,672,000.00 *gal*
Application event totals		1.70	0.00	0.00	303.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.52	0.00	0.00	126.18	3,360,000.00 *gal*
Irrigation Wells	Ground water	0.58	0.00	0.00	28.04	3,780,000.00 *gal*
Application event totals		1.10	0.00	0.00	154.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

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Field 19 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.45	0.00	0.00	109.95	2,928,000.00 *gal*
Irrigation Wells	Ground water	0.51	0.00	0.00	24.43	3,294,000.00 *gal*
Application event totals		0.96	0.00	0.00	134.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.42	0.00	0.00	100.94	2,688,000.00 *gal*
Irrigation Wells	Ground water	0.47	0.00	0.00	22.43	3,024,000.00 *gal*
Application event totals		0.88	0.00	0.00	123.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.54	0.00	0.00	131.58	3,504,000.00 *gal*
Irrigation Wells	Ground water	0.61	0.00	0.00	29.24	3,942,000.00 *gal*
Application event totals		1.15	0.00	0.00	160.82	

Field 2 - 10/15/2022: Rye Silage

Field name: Field 2

Crop: Rye Silage

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.79	14.49	132.74	748.74	1,152,000.00 *gal*
Irrigation Wells	Ground water	0.71	0.00	0.00	141.74	4,416,000.00 *gal*
Irrigation Wells	Ground water	13.71	0.00	0.00	493.00	3,072,000.00 *gal*
Application event totals		139.20	14.49	132.74	1,383.47	

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Field 2 - 10/15/2022: Rye Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.19	12.68	116.15	655.15	1,008,000.00 *gal*
Irrigation Wells	Ground water	0.62	0.00	0.00	124.02	3,864,000.00 *gal*
Irrigation Wells	Ground water	11.99	0.00	0.00	431.37	2,688,000.00 *gal*
Application event totals		121.80	12.68	116.15	1,210.54	

Field 2 - 05/19/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.57	6.07	110.16	588.82	1,008,000.00 *gal*
Irrigation Wells	Ground water	1.37	0.00	0.00	274.61	8,556,000.00 *gal*
Irrigation Wells	Ground water	0.60	0.00	0.00	211.93	3,720,000.00 *gal*
Application event totals		76.54	6.07	110.16	1,075.37	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.80	0.00	0.00	159.45	4,968,000.00 *gal*
Irrigation Wells	Ground water	0.35	0.00	0.00	123.06	2,160,000.00 *gal*
Application event totals		1.14	0.00	0.00	282.51	

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Field 2 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.11	4.81	103.47	565.45	810,000.00 *gal*
Irrigation Wells	Ground water	0.75	0.00	0.00	150.60	4,692,000.00 *gal*
Irrigation Wells	Ground water	14.56	0.00	0.00	523.81	3,264,000.00 *gal*
Application event totals		73.42	4.81	103.47	1,239.86	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.62	0.00	0.00	124.02	3,864,000.00 *gal*
Irrigation Wells	Ground water	0.27	0.00	0.00	95.71	1,680,000.00 *gal*
Application event totals		0.89	0.00	0.00	219.73	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.82	0.00	0.00	163.88	5,106,000.00 *gal*
Irrigation Wells	Ground water	0.36	0.00	0.00	126.47	2,220,000.00 *gal*
Application event totals		1.18	0.00	0.00	290.36	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.98	5.13	110.37	603.15	864,000.00 *gal*
Irrigation Wells	Ground water	0.86	0.00	0.00	172.74	5,382,000.00 *gal*
Application event totals		62.84	5.13	110.37	775.89	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.89	0.00	0.00	177.17	5,520,000.00 *gal*
Irrigation Wells	Ground water	0.39	0.00	0.00	136.73	2,400,000.00 *gal*
Application event totals		1.27	0.00	0.00	313.90	

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Field 20 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 20

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	158.01	43.92	68.32	142.21	161,500.00 *gal*
Application event totals		158.01	43.92	68.32	142.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.98	0.00	0.00	179.11	3,420,000.00 *gal*
Irrigation Wells	Ground water	0.49	0.00	0.00	91.03	1,710,000.00 *gal*
Application event totals		1.48	0.00	0.00	270.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.28	0.00	0.00	51.11	960,000.00 *gal*
Irrigation Wells	Ground water	0.55	0.00	0.00	134.26	1,920,000.00 *gal*
Application event totals		0.83	0.00	0.00	185.36	

Field 20 - 06/15/2023: Corn, silage

Field name: Field 20

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	205.74	31.89	206.77	0.00	660.00 *ton*
Application event totals		205.74	31.89	206.77	0.00	

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Field 20 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.86	0.00	0.00	209.78	3,000,000.00 *gal*
Irrigation Wells	Ground water	0.78	0.00	0.00	37.29	2,700,000.00 *gal*
Application event totals		1.64	0.00	0.00	247.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.66	0.00	0.00	161.11	2,304,000.00 *gal*
Irrigation Wells	Ground water	0.75	0.00	0.00	35.80	2,592,000.00 *gal*
Application event totals		1.41	0.00	0.00	196.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.28	0.00	0.00	67.13	960,000.00 *gal*
Irrigation Wells	Ground water	0.31	0.00	0.00	14.92	1,080,000.00 *gal*
Application event totals		0.59	0.00	0.00	82.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.41	0.00	0.00	100.69	1,440,000.00 *gal*
Irrigation Wells	Ground water	0.47	0.00	0.00	22.38	1,620,000.00 *gal*
Application event totals		0.88	0.00	0.00	123.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

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Field 20 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.58	0.00	0.00	140.97	2,016,000.00 *gal*
Irrigation Wells	Ground water	0.65	0.00	0.00	31.33	2,268,000.00 *gal*
Application event totals		1.23	0.00	0.00	172.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.70	0.00	0.00	171.18	2,448,000.00 *gal*
Irrigation Wells	Ground water	0.79	0.00	0.00	38.04	2,754,000.00 *gal*
Application event totals		1.50	0.00	0.00	209.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.54	0.00	0.00	130.90	1,872,000.00 *gal*
Irrigation Wells	Ground water	0.61	0.00	0.00	29.09	2,106,000.00 *gal*
Application event totals		1.14	0.00	0.00	159.99	

Field 21 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 21

Crop: Wheat, silage, soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	135.06	37.54	58.39	121.55	119,000.00 *gal*
Application event totals		135.06	37.54	58.39	121.55	

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Field 21 - 10/21/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.92	0.00	0.00	167.67	2,760,000.00 *gal*
Irrigation Wells	Ground water	0.46	0.00	0.00	85.22	1,380,000.00 *gal*
Application event totals		1.38	0.00	0.00	252.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.68	0.00	0.00	123.93	2,040,000.00 *gal*
Irrigation Wells	Ground water	0.34	0.00	0.00	62.99	1,020,000.00 *gal*
Application event totals		1.02	0.00	0.00	186.92	

Field 21 - 06/15/2023: Corn, silage

Field name: Field 21

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	231.42	35.87	232.58	0.00	640.00 *ton*
Application event totals		231.42	35.87	232.58	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.92	0.00	0.00	223.87	2,760,000.00 *gal*
Irrigation Wells	Ground water	0.83	0.00	0.00	39.80	2,484,000.00 *gal*
Application event totals		1.75	0.00	0.00	263.67	

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Field 21 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.62	0.00	0.00	151.84	1,872,000.00 *gal*
Irrigation Wells	Ground water	0.70	0.00	0.00	33.74	2,106,000.00 *gal*
Application event totals		1.33	0.00	0.00	185.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.50	0.00	0.00	120.70	1,488,000.00 *gal*
Irrigation Wells	Ground water	0.56	0.00	0.00	26.82	1,674,000.00 *gal*
Application event totals		1.06	0.00	0.00	147.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.67	0.00	0.00	163.52	2,016,000.00 *gal*
Irrigation Wells	Ground water	0.76	0.00	0.00	36.34	2,268,000.00 *gal*
Application event totals		1.43	0.00	0.00	199.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.58	0.00	0.00	140.16	1,728,000.00 *gal*
Irrigation Wells	Ground water	0.65	0.00	0.00	31.15	1,944,000.00 *gal*
Application event totals		1.23	0.00	0.00	171.31	

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Field 21 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.46	0.00	0.00	112.91	1,392,000.00 *gal*
Irrigation Wells	Ground water	0.52	0.00	0.00	25.09	1,566,000.00 *gal*
Application event totals		0.99	0.00	0.00	138.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.56	0.00	0.00	136.27	1,680,000.00 *gal*
Irrigation Wells	Ground water	0.63	0.00	0.00	30.28	1,890,000.00 *gal*
Application event totals		1.19	0.00	0.00	166.55	

Field 29 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 29

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.67	4.16	69.96	393.47	391,000.00 *gal*
Irrigation Wells	Ground water	0.68	0.00	0.00	0.00	2,760,000.00 *gal*
Application event totals		44.34	4.16	69.96	393.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.71	0.00	0.00	0.00	2,880,000.00 *gal*
Application event totals		0.71	0.00	0.00	0.00	

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Field 29 - 10/22/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.45	10.74	98.33	554.67	558,000.00 *gal*
Irrigation Wells	Ground water	0.45	0.00	0.00	89.54	1,824,000.00 *gal*
Application event totals		92.89	10.74	98.33	644.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.60	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		0.60	0.00	0.00	0.00	

Field 29 - 07/01/2023: Corn, silage

Field name: Field 29

Crop: Corn, silage

Plant date: 07/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.25	5.63	102.29	546.76	612,000.00 *gal*
Irrigation Wells	Ground water	1.40	0.00	0.00	0.00	5,700,000.00 *gal*
Application event totals		70.65	5.63	102.29	546.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.30	0.00	0.00	0.00	5,280,000.00 *gal*
Application event totals		1.30	0.00	0.00	0.00	

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Field 29 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.19	0.00	0.00	0.00	4,860,000.00 *gal*
Application event totals		1.19	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.33	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		1.33	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.04	6.21	133.63	730.29	684,000.00 *gal*
Irrigation Wells	Ground water	1.31	0.00	0.00	0.00	5,340,000.00 *gal*
Application event totals		76.35	6.21	133.63	730.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.24	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		1.24	0.00	0.00	0.00	

Field 3 - 09/11/2022: Wheat, silage, soft dough

Field name: Field 3

Crop: Wheat, silage, soft dough

Plant date: 09/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 3 - 09/11/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	116.91	11.15	187.31	1,053.45	1,170,000.00 *gal*
Irrigation Wells	Ground water	1.64	0.00	0.00	327.30	7,452,000.00 *gal*
Application event totals		118.54	11.15	187.31	1,380.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	155.65	18.08	165.56	933.87	1,050,000.00 *gal*
Irrigation Wells	Ground water	1.06	0.00	0.00	212.14	4,830,000.00 *gal*
Irrigation Wells	Ground water	0.46	0.00	0.00	163.72	2,100,000.00 *gal*
Application event totals		157.17	18.08	165.56	1,309.73	

Field 3 - 05/26/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	153.07	12.45	226.12	1,208.64	1,512,000.00 *gal*
Irrigation Wells	Ground water	1.27	0.00	0.00	254.57	5,796,000.00 *gal*
Application event totals		154.34	12.45	226.12	1,463.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.15	0.00	0.00	230.32	5,244,000.00 *gal*
Application event totals		1.15	0.00	0.00	230.32	

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Field 3 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.15	0.00	0.00	230.32	5,244,000.00 *gal*
Application event totals		1.15	0.00	0.00	230.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.09	0.00	0.00	218.20	4,968,000.00 *gal*
Application event totals		1.09	0.00	0.00	218.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	116.62	9.65	207.67	1,134.88	1,188,000.00 *gal*
Irrigation Wells	Ground water	1.00	0.00	0.00	200.02	4,554,000.00 *gal*
Irrigation Wells	Ground water	0.77	0.00	0.00	149.40	3,488,760.00 *gal*
Application event totals		118.38	9.65	207.67	1,484.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.03	0.00	0.00	206.08	4,692,000.00 *gal*
Irrigation Wells	Ground water	0.45	0.00	0.00	159.04	2,040,000.00 *gal*
Application event totals		1.48	0.00	0.00	365.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.76	0.00	0.00	151.53	3,450,000.00 *gal*
Irrigation Wells	Ground water	0.33	0.00	0.00	116.94	1,500,000.00 *gal*
Application event totals		1.09	0.00	0.00	268.47	

Field 30 - 10/22/2022: Wheat, silage, soft dough

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Field 30 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 30

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.97	6.67	112.11	630.52	774,000.00 *gal*
Irrigation Wells	Ground water	0.86	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		70.83	6.67	112.11	630.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.84	8.69	79.60	449.02	558,000.00 *gal*
Irrigation Wells	Ground water	0.86	0.00	0.00	172.74	4,347,000.00 *gal*
Application event totals		75.70	8.69	79.60	621.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.70	0.00	0.00	0.00	3,540,000.00 *gal*
Application event totals		0.70	0.00	0.00	0.00	

Field 30 - 07/01/2023: Corn, silage

Field name: Field 30

Crop: Corn, silage

Plant date: 07/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.87	8.45	153.44	820.15	810,000.00 *gal*
Irrigation Wells	Ground water	1.55	0.00	0.00	0.00	5,580,000.00 *gal*
Application event totals		105.42	8.45	153.44	820.15	

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Field 30 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.44	0.00	0.00	0.00	5,160,000.00 *gal*
Application event totals		1.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.37	0.00	0.00	0.00	4,920,000.00 *gal*
Application event totals		1.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.76	7.60	163.41	893.00	738,000.00 *gal*
Irrigation Wells	Ground water	1.47	0.00	0.00	0.00	5,280,000.00 *gal*
Application event totals		93.23	7.60	163.41	893.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.59	0.00	0.00	0.00	5,700,000.00 *gal*
Application event totals		1.59	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.40	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		1.40	0.00	0.00	0.00	

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Field 31 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 31

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	158.99	38.95	250.40	0.00	176.00 *ton*
Application event totals		158.99	38.95	250.40	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.06	0.00	0.00	0.00	4,560,000.00 *gal*
Application event totals		1.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.00	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		1.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.86	9.16	83.88	473.16	504,000.00 *gal*
Irrigation Wells	Ground water	0.93	0.00	0.00	0.00	4,020,000.00 *gal*
Application event totals		79.79	9.16	83.88	473.16	

Field 31 - 05/15/2023: Corn, silage

Field name: Field 31

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 31 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	160.65	13.07	237.32	1,268.49	1,044,000.00 *gal*
Irrigation Wells	Ground water	1.46	0.00	0.00	0.00	4,380,000.00 *gal*
Application event totals		162.11	13.07	237.32	1,268.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.18	0.00	0.00	0.00	3,540,000.00 *gal*
Application event totals		1.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.38	0.00	0.00	0.00	4,140,000.00 *gal*
Application event totals		1.38	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.43	8.89	191.31	1,045.46	720,000.00 *gal*
Irrigation Wells	Ground water	1.02	0.00	0.00	0.00	3,060,000.00 *gal*
Application event totals		108.45	8.89	191.31	1,045.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.00	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		1.00	0.00	0.00	0.00	

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Field 31 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.90	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.90	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.98	0.00	0.00	0.00	2,940,000.00 *gal*
Application event totals		0.98	0.00	0.00	0.00	

Field 32 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 32

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	152.29	37.31	239.85	0.00	192.00 *ton*
Application event totals		152.29	37.31	239.85	0.00	

11/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.83	0.00	0.00	0.00	4,080,000.00 *gal*
Application event totals		0.83	0.00	0.00	0.00	

01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.01	0.00	0.00	0.00	4,980,000.00 *gal*
Application event totals		1.01	0.00	0.00	0.00	

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Field 32 - 10/22/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.50	17.71	162.21	915.00	1,110,000.00 *gal*
Irrigation Wells	Ground water	0.90	0.00	0.00	0.00	4,440,000.00 *gal*
Application event totals		153.40	17.71	162.21	915.00	

Field 32 - 05/15/2023: Corn, silage

Field name: Field 32

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.49	8.82	160.26	856.60	705,000.00 *gal*
Irrigation Wells	Ground water	1.74	0.00	0.00	0.00	5,220,000.00 *gal*
Application event totals		110.23	8.82	160.26	856.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.32	0.00	0.00	0.00	3,960,000.00 *gal*
Application event totals		1.32	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.05	7.04	151.45	827.66	570,000.00 *gal*
Irrigation Wells	Ground water	1.16	0.00	0.00	0.00	3,480,000.00 *gal*
Application event totals		86.21	7.04	151.45	827.66	

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Field 32 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.86	0.00	0.00	0.00	2,580,000.00 *gal*
Application event totals		0.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.92	0.00	0.00	0.00	2,760,000.00 *gal*
Application event totals		0.92	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.90	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.90	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.81	6.85	147.47	805.88	555,000.00 *gal*
Irrigation Wells	Ground water	1.00	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		83.81	6.85	147.47	805.88	

Field 4 - 09/11/2022: Rye Silage

Field name: Field 4

Crop: Rye Silage

Plant date: 09/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 4 - 09/11/2022: Rye Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	284.86	33.09	303.01	1,709.18	1,770,000.00 *gal*
Irrigation Wells	Ground water	1.77	0.00	0.00	344.81	7,416,360.00 *gal*
Application event totals		286.63	33.09	303.01	2,053.99	

Field 4 - 04/24/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 04/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	131.90	10.73	194.84	1,041.46	1,200,000.00 *gal*
Irrigation Wells	Ground water	1.11	0.00	0.00	216.27	4,651,680.00 *gal*
Application event totals		133.01	10.73	194.84	1,257.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.12	0.00	0.00	218.73	4,704,540.00 *gal*
Application event totals		1.12	0.00	0.00	218.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.13	0.00	0.00	221.19	4,757,400.00 *gal*
Application event totals		1.13	0.00	0.00	221.19	

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Field 4 - 04/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.98	0.00	0.00	191.70	4,123,080.00 *gal*
Application event totals		0.98	0.00	0.00	191.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.18	0.00	0.00	231.02	4,968,840.00 *gal*
Application event totals		1.18	0.00	0.00	231.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.50	10.06	216.36	1,182.37	1,140,000.00 *gal*
Irrigation Wells	Ground water	1.26	0.00	0.00	245.77	5,286,000.00 *gal*
Application event totals		122.76	10.06	216.36	1,428.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.20	0.00	0.00	233.48	5,021,700.00 *gal*
Application event totals		1.20	0.00	0.00	233.48	

Field 5 - 09/11/2022: Rye Silage

Field name: Field 5

Crop: Rye Silage

Plant date: 09/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 5 - 09/11/2022: Rye Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	239.65	27.84	254.92	1,437.92	1,404,000.00 *gal*
Irrigation Wells	Ground water	1.44	0.00	0.00	281.51	5,708,880.00 *gal*
Application event totals		241.10	27.84	254.92	1,719.43	

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.69	11.61	210.79	1,126.67	1,224,000.00 *gal*
Irrigation Wells	Ground water	0.52	0.00	0.00	183.13	2,040,000.00 *gal*
Application event totals		143.21	11.61	210.79	1,309.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.98	0.00	0.00	195.42	3,864,000.00 *gal*
Irrigation Wells	Ground water	18.90	0.00	0.00	679.74	2,688,000.00 *gal*
Application event totals		19.87	0.00	0.00	875.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.08	0.00	0.00	216.36	4,278,000.00 *gal*
Irrigation Wells	Ground water	0.47	0.00	0.00	166.98	1,860,000.00 *gal*
Application event totals		1.55	0.00	0.00	383.34	

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Field 5 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.22	10.78	231.89	1,267.23	1,152,000.00 *gal*
Irrigation Wells	Ground water	0.86	0.00	0.00	166.82	3,383,040.00 *gal*
Application event totals		131.07	10.78	231.89	1,434.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.98	0.00	0.00	190.28	3,858,780.00 *gal*
Application event totals		0.98	0.00	0.00	190.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.12	0.00	0.00	218.95	4,440,240.00 *gal*
Application event totals		1.12	0.00	0.00	218.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.07	0.00	0.00	208.53	4,228,800.00 *gal*
Application event totals		1.07	0.00	0.00	208.53	

Field 6 - 11/01/2021: Alfalfa, hay

Field name: Field 6

Crop: Alfalfa, hay

Plant date: 11/01/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

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Field 6 - 11/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.68	9.13	153.30	862.21	1,260,000.00 *gal*
Irrigation Wells	Ground water	0.97	0.00	0.00	193.47	5,796,000.00 *gal*
Irrigation Wells	Ground water	18.71	0.00	0.00	672.94	4,032,000.00 *gal*
Application event totals		115.36	9.13	153.30	1,728.62	

02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.77	0.00	0.00	149.67	4,598,820.00 *gal*
Application event totals		0.77	0.00	0.00	149.67	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.33	8.73	158.55	847.48	1,395,000.00 *gal*
Irrigation Wells	Ground water	1.07	0.00	0.00	214.20	6,417,000.00 *gal*
Application event totals		108.40	8.73	158.55	1,061.68	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.85	0.00	0.00	170.44	5,106,000.00 *gal*
Application event totals		0.85	0.00	0.00	170.44	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.88	0.00	0.00	175.04	5,244,000.00 *gal*
Application event totals		0.88	0.00	0.00	175.04	

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Field 6 - 11/01/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.77	0.00	0.00	154.32	4,623,000.00 *gal*
Application event totals		0.77	0.00	0.00	154.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.77	0.00	0.00	154.32	4,623,000.00 *gal*
Application event totals		0.77	0.00	0.00	154.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.45	6.58	141.49	773.21	1,065,000.00 *gal*
Irrigation Wells	Ground water	0.63	0.00	0.00	122.15	3,753,060.00 *gal*
Application event totals		80.08	6.58	141.49	895.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.99	0.00	0.00	198.08	5,934,000.00 *gal*
Application event totals		0.99	0.00	0.00	198.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	0.79	0.00	0.00	154.83	4,757,400.00 *gal*
Application event totals		0.79	0.00	0.00	154.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.55	5.09	109.60	598.96	825,000.00 *gal*
Irrigation Wells	Ground water	0.63	0.00	0.00	126.68	3,795,000.00 *gal*
Application event totals		62.18	5.09	109.60	725.64	

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Field 7 - 10/15/2022: Wheat, silage, soft dough

Field name: Field 7

Crop: Wheat, silage, soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.33	11.38	191.20	1,075.31	1,320,000.00 *gal*
Irrigation Wells	Ground water	1.13	0.00	0.00	221.19	5,708,880.00 *gal*
Application event totals		120.47	11.38	191.20	1,296.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	181.06	21.03	192.59	1,086.34	1,350,000.00 *gal*
Irrigation Wells	Ground water	1.47	0.00	0.00	286.73	7,400,400.00 *gal*
Application event totals		182.53	21.03	192.59	1,373.07	

Field 7 - 06/08/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.67	8.27	150.19	802.79	1,110,000.00 *gal*
Irrigation Wells	Ground water	1.32	0.00	0.00	263.23	6,624,000.00 *gal*
Application event totals		102.99	8.27	150.19	1,066.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.21	0.00	0.00	235.52	6,078,900.00 *gal*
Application event totals		1.21	0.00	0.00	235.52	

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Field 7 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.31	0.00	0.00	256.01	6,607,500.00 *gal*
Application event totals		1.31	0.00	0.00	256.01	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.39	0.00	0.00	270.34	6,977,520.00 *gal*
Application event totals		1.39	0.00	0.00	270.34	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.56	10.81	232.49	1,270.53	1,470,000.00 *gal*
Irrigation Wells	Ground water	1.34	0.00	0.00	262.15	6,766,080.00 *gal*
Application event totals		131.90	10.81	232.49	1,532.68	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.20	0.00	0.00	233.48	6,026,040.00 *gal*
Application event totals		1.20	0.00	0.00	233.48	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Wells	Ground water	1.32	0.00	0.00	258.05	6,660,360.00 *gal*
Application event totals		1.32	0.00	0.00	258.05	

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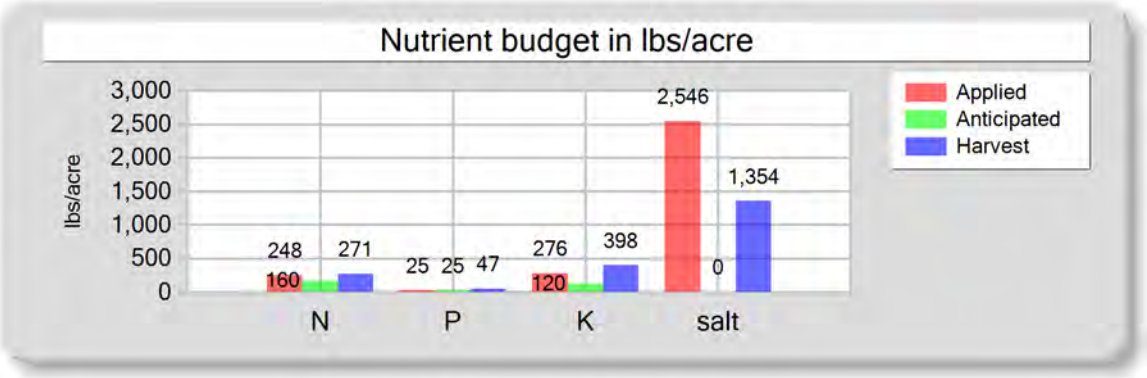
B. NUTRIENT BUDGET

Field 1 - 10/15/2022: Wheat, silage, soft dough

Field name: Field 1 Crop: Wheat, silage, soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.47	24.62	276.45	1,557.53
Fresh water	16.18	0.00	0.00	988.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.66	24.62	276.45	2,545.84
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	271.16	47.00	397.70	1,354.48
Nutrient balance	-23.50	-22.38	-121.24	1,191.36
Applied to removed ratio	0.91	0.52	0.70	1.88

Fresh water applied	
12,554,040.00	*gallons*
462.32	*acre-inches*
13.21	*inches/acre*

Process wastewater applied	
1,605,000.00	*gallons*
59.11	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

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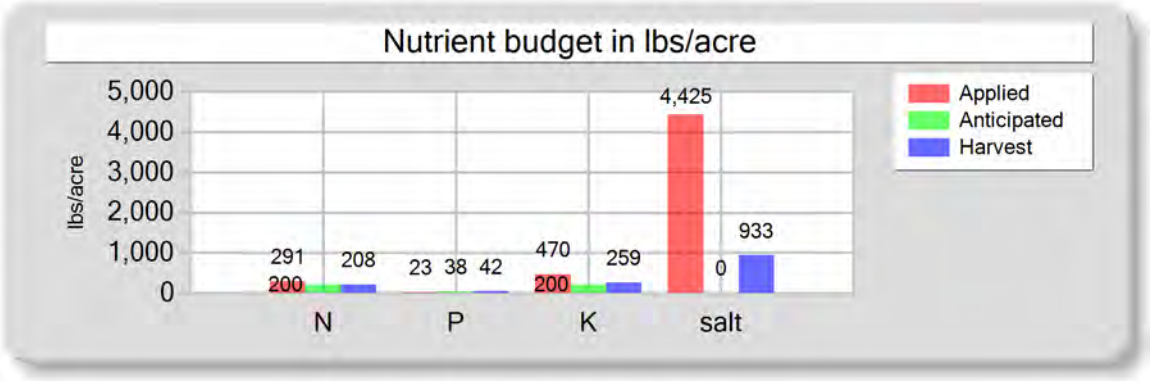
Field 1 - 05/26/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.42	22.70	469.95	2,556.22
Fresh water	8.54	0.00	0.00	1,868.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.96	22.70	469.95	4,424.79
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	207.77	41.55	258.56	933.40
Nutrient balance	83.19	-18.85	211.39	3,491.39
Applied to removed ratio	1.40	0.55	1.82	4.74

Fresh water applied	
35,814,000.00	*gallons*
1,318.91	*acre-inches*
37.68	*inches/acre*

Process wastewater applied	
2,565,000.00	*gallons*
94.46	*acre-inches*
2.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

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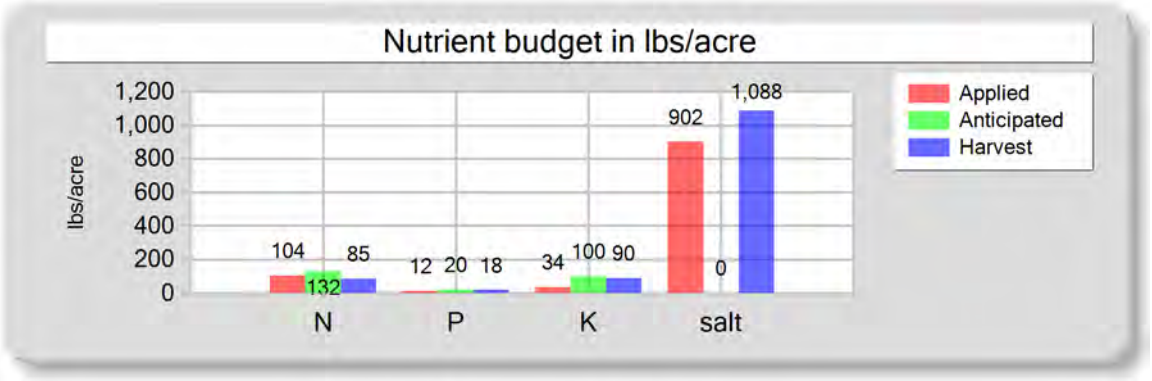
Field 10-11 - 10/26/2022: Wheat Hay

Field name: Field 10-11 Crop: Wheat Hay Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	94.16	11.99	33.59	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.65	0.00	0.00	901.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	103.80	11.99	33.59	901.56
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	85.14	18.11	89.66	1,088.19
Nutrient balance	18.67	-6.12	-56.08	-186.63
Applied to removed ratio	1.22	0.66	0.37	0.83

Fresh water applied	
47,880,000.00	*gallons*
1,763.26	*acre-inches*
11.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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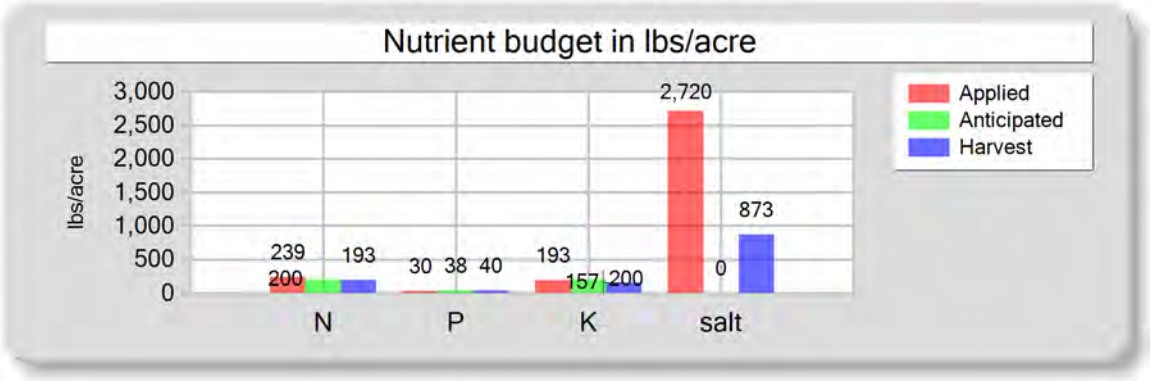
Field 10-11 - 06/18/2023: Corn, silage

Field name: Field 10-11 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	191.58	29.69	192.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.97	0.00	0.00	2,719.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.55	29.69	192.53	2,719.75
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	192.94	40.38	157.04	873.06
Nutrient balance	45.61	-10.69	35.49	1,846.69
Applied to removed ratio	1.24	0.74	1.23	3.12

Fresh water applied	
144,252,000.00	*gallons*
5,312.31	*acre-inches*
35.18	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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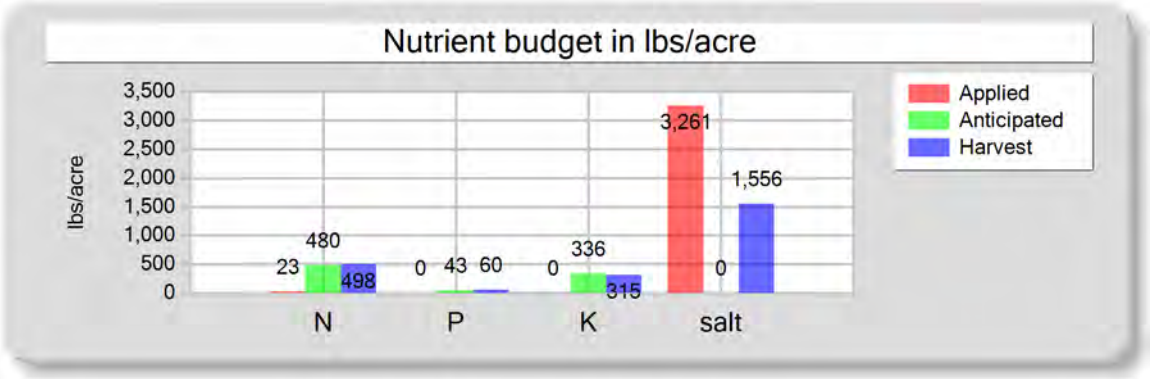
Field 12 - 10/01/2020: Alfalfa, hay

Field name: Field 12 Crop: Alfalfa, hay Plant date: 10/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.45	0.00	0.00	3,261.36
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	23.45	0.00	0.00	3,261.36
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	498.18	59.70	314.97	1,556.31
Nutrient balance	-474.73	-59.70	-314.97	1,705.05
Applied to removed ratio	0.05	0.00	0.00	2.10

Fresh water applied	
84,960,000.00	*gallons*
3,128.79	*acre-inches*
41.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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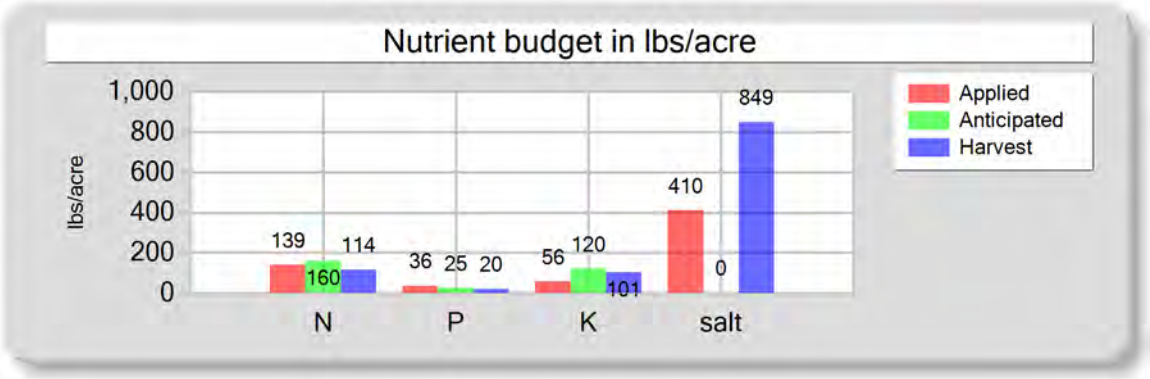
Field 15 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 15 Crop: Wheat, silage, soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	130.36	36.23	56.36	117.33
Fresh water	1.60	0.00	0.00	292.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.96	36.23	56.36	409.70
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	114.41	20.19	100.95	848.89
Nutrient balance	24.56	16.04	-44.58	-439.18
Applied to removed ratio	1.21	1.79	0.56	0.48

Fresh water applied	
21,300,000.00	*gallons*
784.41	*acre-inches*
7.07	*inches/acre*

Process wastewater applied	
510,000.00	*gallons*
18.78	*acre-inches*
0.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

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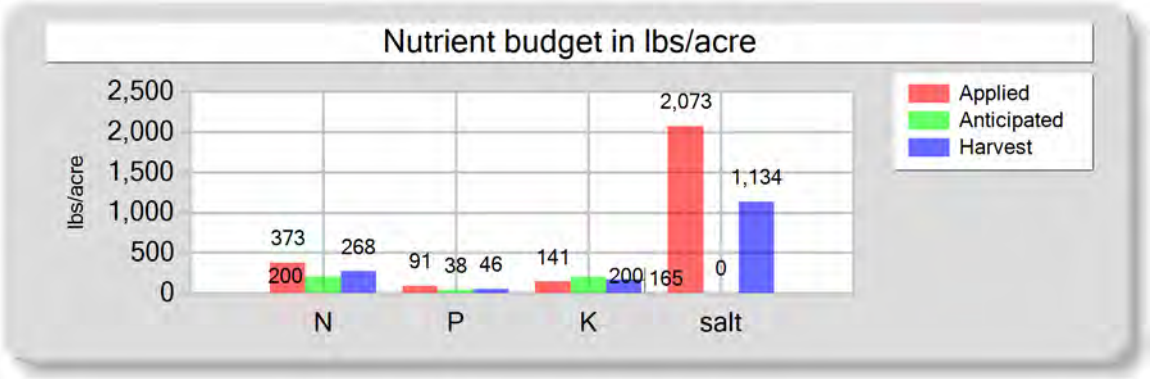
Field 15 - 05/31/2023: Corn, silage

Field name: Field 15 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	325.91	90.58	140.91	293.32
Fresh water	7.84	0.00	0.00	1,780.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	372.75	90.58	140.91	2,073.46
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	267.73	46.34	164.76	1,133.73
Nutrient balance	105.02	44.24	-23.85	939.73
Applied to removed ratio	1.39	1.95	0.86	1.83

Fresh water applied	
104,310,000.00	*gallons*
3,841.38	*acre-inches*
34.61	*inches/acre*

Process wastewater applied	
1,275,000.00	*gallons*
46.95	*acre-inches*
0.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

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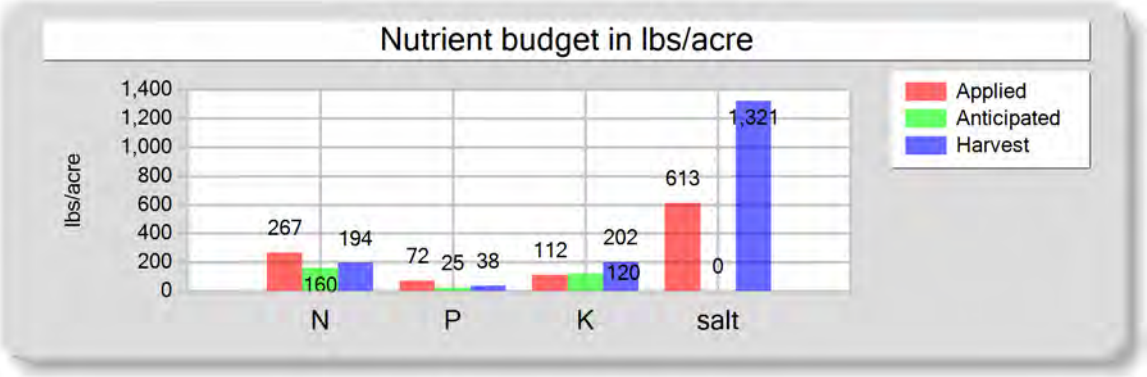
Field 16 - 10/20/2022: Wheat, silage, soft dough

Field name: Field 16 Crop: Wheat, silage, soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.40	71.82	111.72	232.56
Fresh water	2.08	0.00	0.00	380.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.48	71.82	111.72	613.47
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	194.46	38.13	202.08	1,321.38
Nutrient balance	73.02	33.69	-90.36	-707.92
Applied to removed ratio	1.38	1.88	0.55	0.46

Fresh water applied	
17,460,000.00	*gallons*
642.99	*acre-inches*
9.19	*inches/acre*

Process wastewater applied	
637,500.00	*gallons*
23.48	*acre-inches*
0.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

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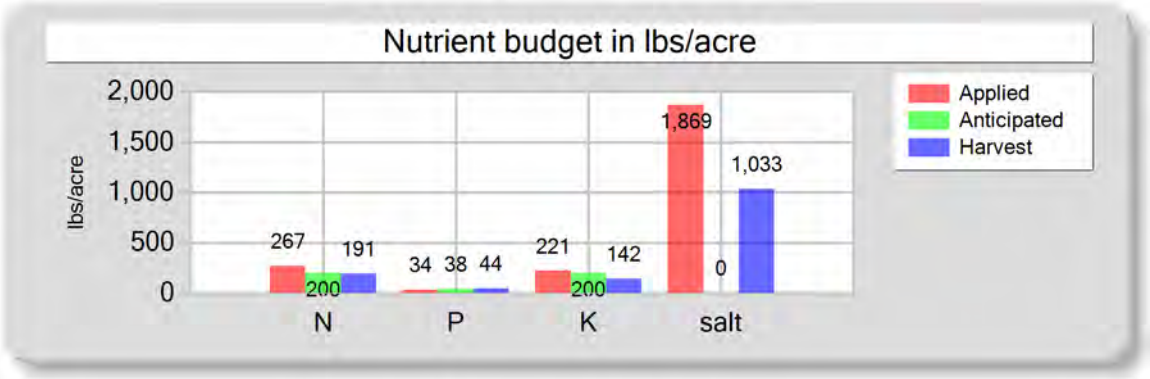
Field 16 - 06/20/2023: Corn, silage

Field name: Field 16 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	219.54	34.03	220.64	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.84	0.00	0.00	1,869.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.38	34.03	220.64	1,869.22
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	190.81	44.03	141.88	1,032.92
Nutrient balance	76.57	-10.00	78.76	836.30
Applied to removed ratio	1.40	0.77	1.56	1.81

Fresh water applied	
74,168,000.00	*gallons*
2,731.36	*acre-inches*
39.02	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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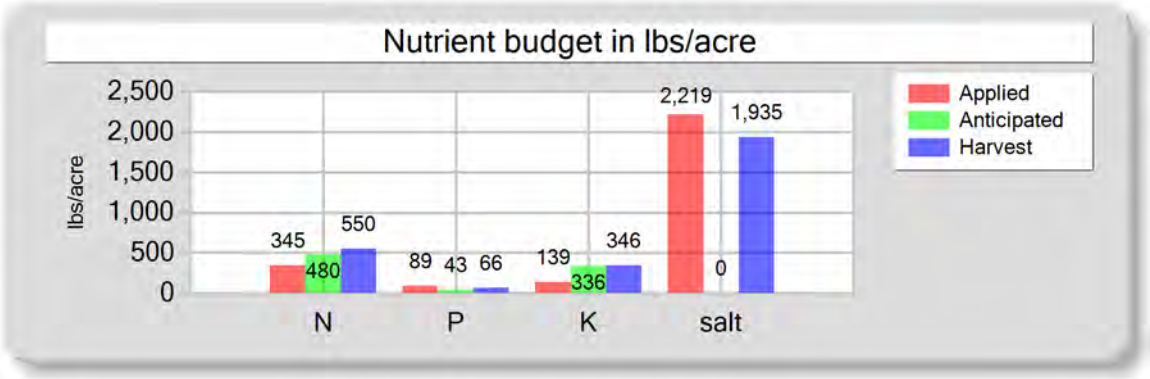
Reporting period 01/01/2023 to 12/31/2023.

Field 17 - 09/25/2018: Alfalfa, hay

Field name: Field 17 Crop: Alfalfa, hay Plant date: 09/25/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	321.56	89.37	139.03	289.40
Fresh water	9.37	0.00	0.00	1,929.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	344.93	89.37	139.03	2,218.63
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	549.87	65.62	346.22	1,934.74
Nutrient balance	-204.94	23.75	-207.19	283.89
Applied to removed ratio	0.63	1.36	0.40	1.15

Fresh water applied	
94,320,000.00	*gallons*
3,473.48	*acre-inches*
41.35	*inches/acre*

Process wastewater applied	
952,000.00	*gallons*
35.06	*acre-inches*
0.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

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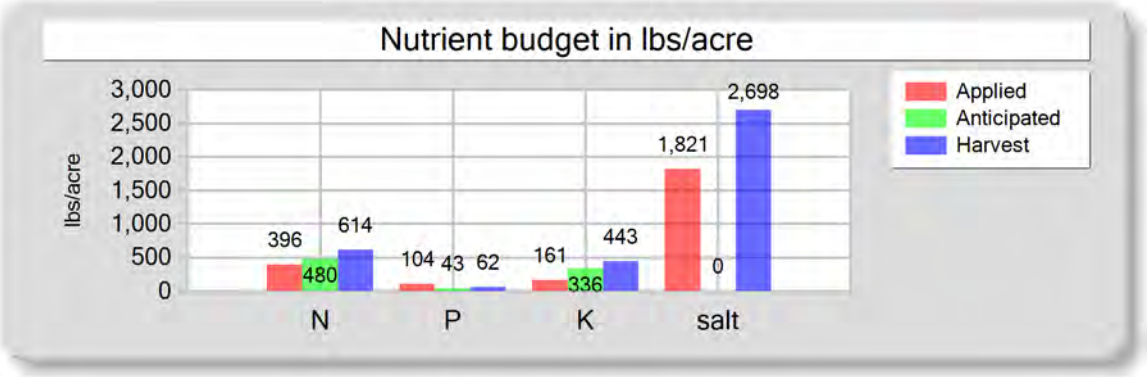
Field 18 - 10/12/2022: Alfalfa, hay

Field name: Field 18 Crop: Alfalfa, hay Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	373.01	103.67	161.27	335.71
Fresh water	9.31	0.00	0.00	1,485.44
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	396.32	103.67	161.27	1,821.15
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	614.40	61.63	442.59	2,697.56
Nutrient balance	-218.08	42.05	-281.32	-876.40
Applied to removed ratio	0.65	1.68	0.36	0.68

Fresh water applied	
83,628,000.00	*gallons*
3,079.73	*acre-inches*
41.06	*inches/acre*

Process wastewater applied	
986,000.00	*gallons*
36.31	*acre-inches*
0.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

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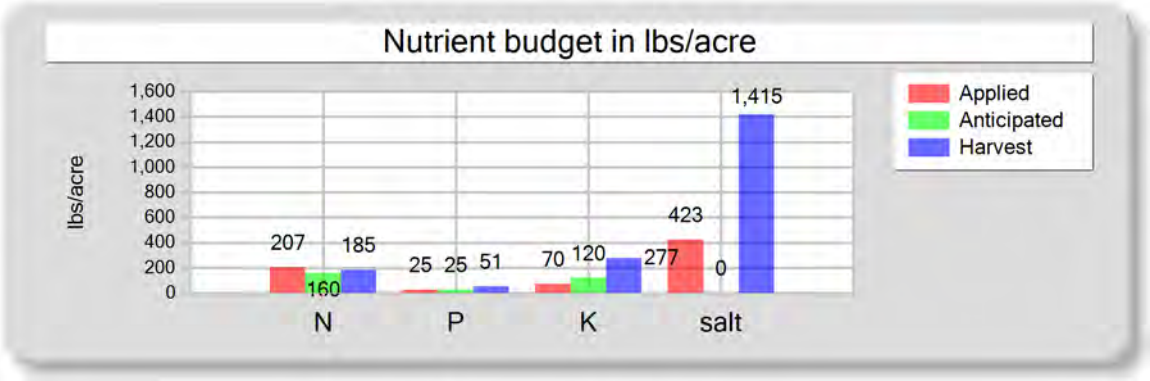
Field 19 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 19 Crop: Wheat, silage, soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	197.47	25.16	70.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.16	0.00	0.00	423.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.63	25.16	70.44	423.17
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	184.64	51.29	276.96	1,415.40
Nutrient balance	21.99	-26.13	-206.52	-992.23
Applied to removed ratio	1.12	0.49	0.25	0.30

Fresh water applied	
13,980,000.00	*gallons*
514.84	*acre-inches*
9.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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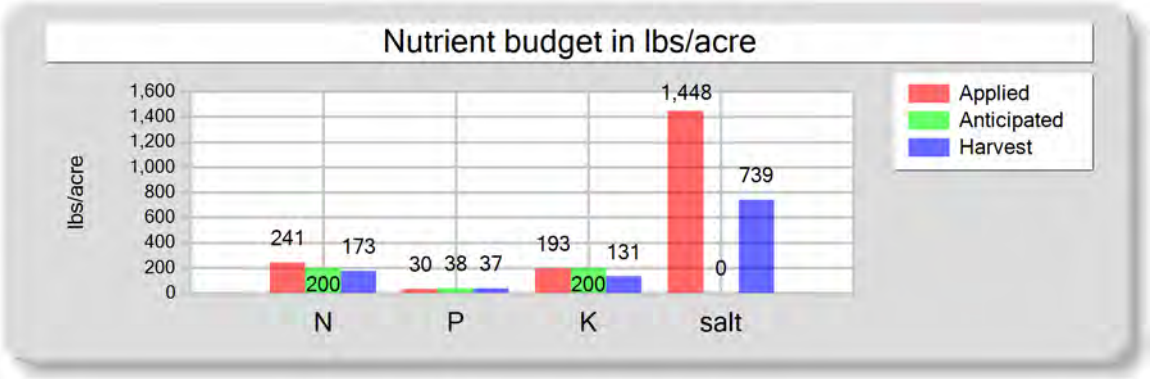
Field 19 - 06/03/2023: Corn, silage

Field name: Field 19 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	192.52	29.84	193.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.39	0.00	0.00	1,447.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.91	29.84	193.48	1,447.92
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	172.52	37.30	130.55	738.51
Nutrient balance	68.39	-7.46	62.93	709.42
Applied to removed ratio	1.40	0.80	1.48	1.96

Fresh water applied	
60,744,000.00	*gallons*
2,236.99	*acre-inches*
41.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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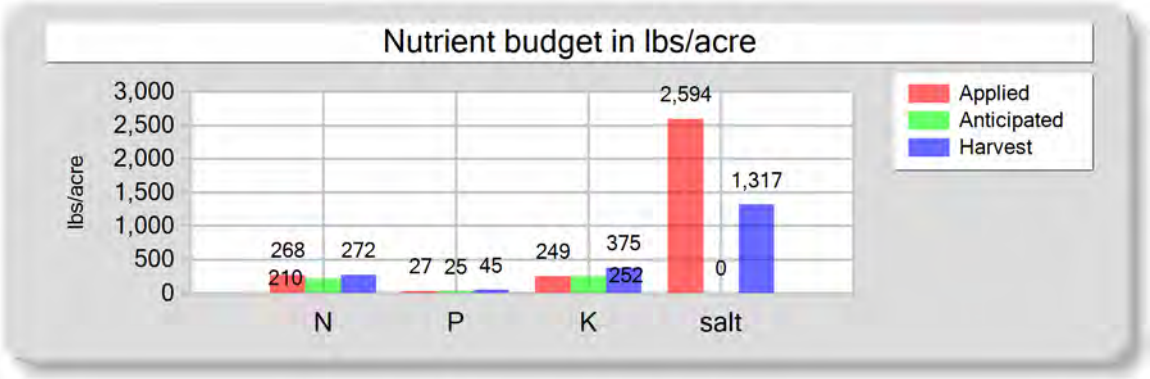
Field 2 - 10/15/2022: Rye Silage

Field name: Field 2 Crop: Rye Silage Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	233.98	27.18	248.89	1,403.89
Fresh water	27.03	0.00	0.00	1,190.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.01	27.18	248.89	2,594.01
Anticipated crop nutrient removal	210.00	25.20	252.00	0.00
Actual crop nutrient removal	272.04	45.34	375.31	1,316.84
Nutrient balance	-4.03	-18.16	-126.42	1,277.17
Applied to removed ratio	0.99	0.60	0.66	1.97

Fresh water applied	
14,040,000.00	*gallons*
517.05	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

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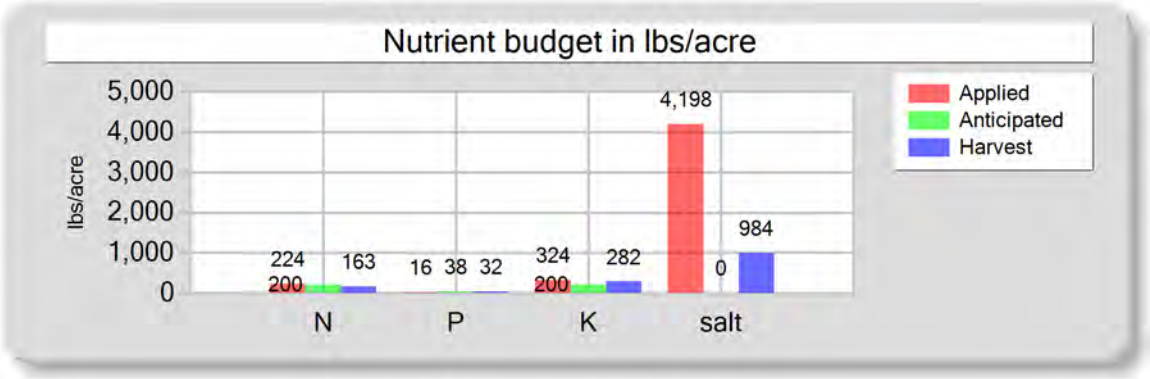
Field 2 - 05/19/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.66	16.01	324.00	1,757.43
Fresh water	22.63	0.00	0.00	2,440.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.29	16.01	324.00	4,197.62
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	163.38	32.03	281.90	984.46
Nutrient balance	60.91	-16.03	42.10	3,213.16
Applied to removed ratio	1.37	0.50	1.15	4.26

Fresh water applied	
53,532,000.00	*gallons*
1,971.40	*acre-inches*
37.91	*inches/acre*

Process wastewater applied	
2,682,000.00	*gallons*
98.77	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

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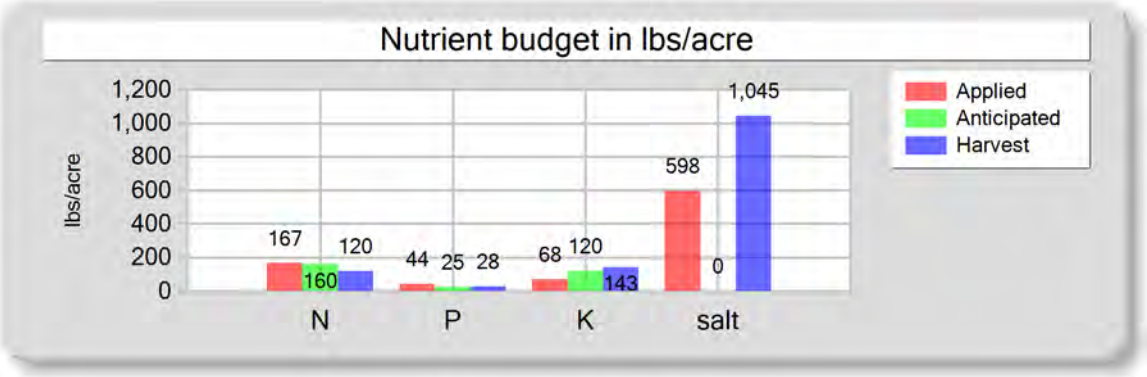
Field 20 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 20 Crop: Wheat, silage, soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	158.01	43.92	68.32	142.21
Fresh water	2.30	0.00	0.00	455.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.31	43.92	68.32	597.71
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	119.70	28.02	142.63	1,044.79
Nutrient balance	47.61	15.90	-74.31	-447.07
Applied to removed ratio	1.40	1.57	0.48	0.57

Fresh water applied	
8,010,000.00	*gallons*
294.98	*acre-inches*
10.17	*inches/acre*

Process wastewater applied	
161,500.00	*gallons*
5.95	*acre-inches*
0.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

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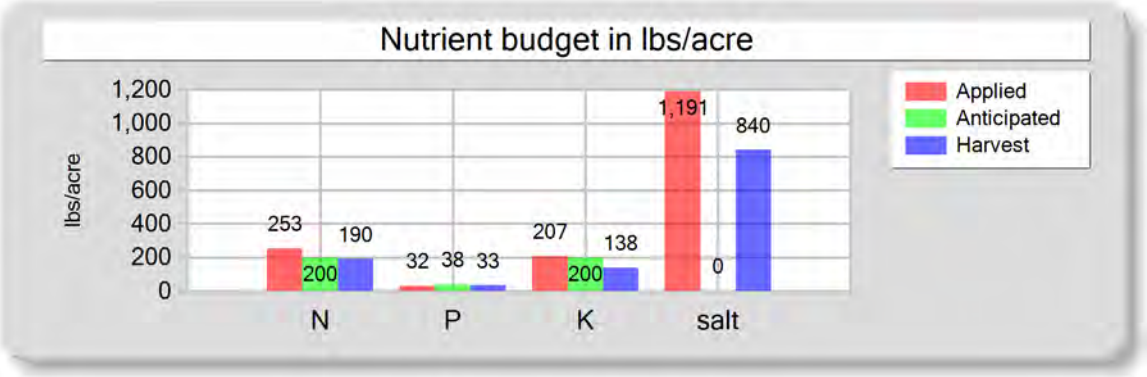
Field 20 - 06/15/2023: Corn, silage

Field name: Field 20 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	205.74	31.89	206.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.39	0.00	0.00	1,190.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.13	31.89	206.77	1,190.60
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	190.01	33.25	137.76	840.29
Nutrient balance	63.12	-1.36	69.01	350.31
Applied to removed ratio	1.33	0.96	1.50	1.42

Fresh water applied	
29,160,000.00	*gallons*
1,073.86	*acre-inches*
37.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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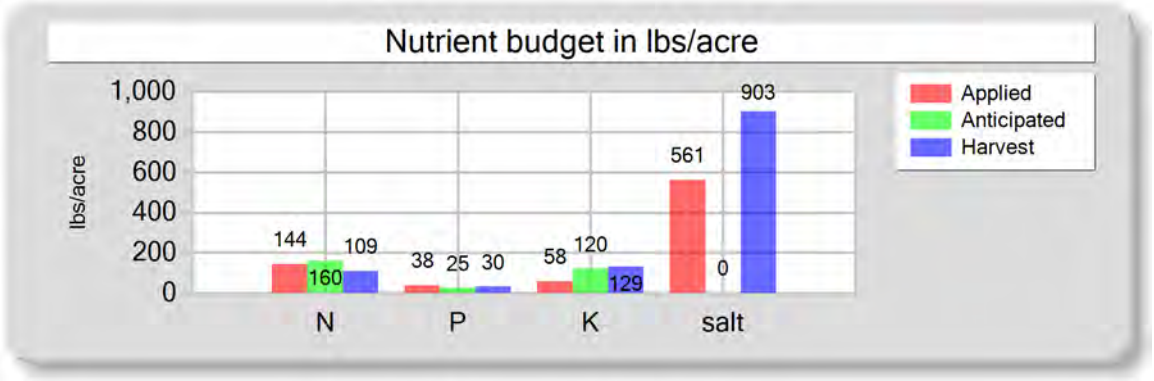
Field 21 - 10/21/2022: Wheat, silage, soft dough

Field name: Field 21 Crop: Wheat, silage, soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	135.06	37.54	58.39	121.55
Fresh water	2.40	0.00	0.00	439.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.46	37.54	58.39	561.36
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	109.08	30.44	129.38	903.20
Nutrient balance	35.38	7.10	-70.99	-341.84
Applied to removed ratio	1.32	1.23	0.45	0.62

Fresh water applied	
7,200,000.00	*gallons*
265.15	*acre-inches*
10.61	*inches/acre*

Process wastewater applied	
119,000.00	*gallons*
4.38	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

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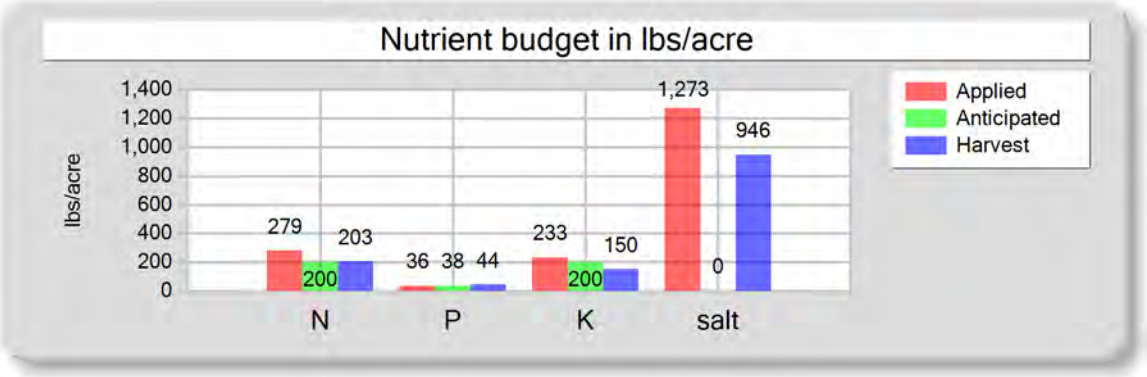
Field 21 - 06/15/2023: Corn, silage

Field name: Field 21 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	231.42	35.87	232.58	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.97	0.00	0.00	1,272.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.39	35.87	232.58	1,272.51
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	203.25	43.55	150.02	945.97
Nutrient balance	76.15	-7.68	82.57	326.54
Applied to removed ratio	1.37	0.82	1.55	1.35

Fresh water applied	
26,868,000.00	*gallons*
989.46	*acre-inches*
39.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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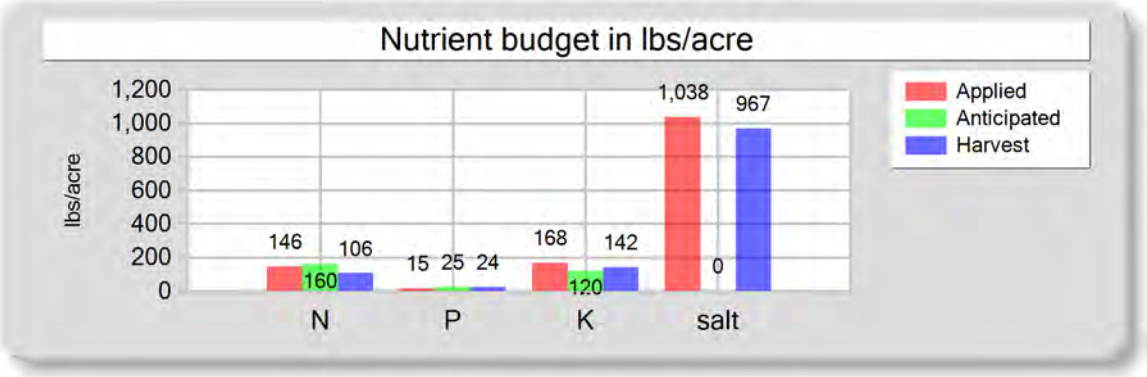
Field 29 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 29 Crop: Wheat, silage, soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	136.11	14.90	168.29	948.14
Fresh water	2.44	0.00	0.00	89.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	145.55	14.90	168.29	1,037.68
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	105.70	24.02	141.74	967.50
Nutrient balance	39.84	-9.12	26.56	70.18
Applied to removed ratio	1.38	0.62	1.19	1.07

Fresh water applied	
9,924,000.00	*gallons*
365.47	*acre-inches*
10.75	*inches/acre*

Process wastewater applied	
949,000.00	*gallons*
34.95	*acre-inches*
1.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

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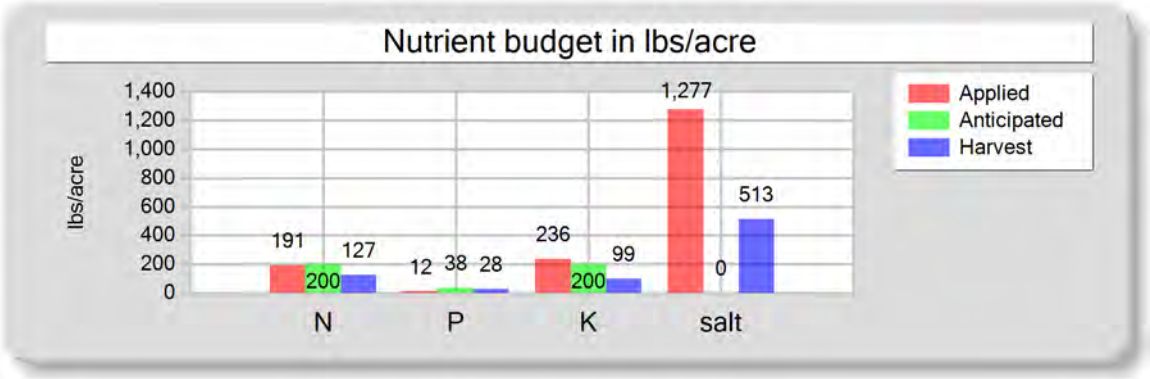
Field 29 - 07/01/2023: Corn, silage

Field name: Field 29 Crop: Corn, silage Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	144.29	11.84	235.93	1,277.05
Fresh water	7.76	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.05	11.84	235.93	1,277.05
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	127.11	28.25	98.86	513.25
Nutrient balance	63.94	-16.40	137.06	763.80
Applied to removed ratio	1.50	0.42	2.39	2.49

Fresh water applied	
31,620,000.00	*gallons*
1,164.46	*acre-inches*
34.25	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

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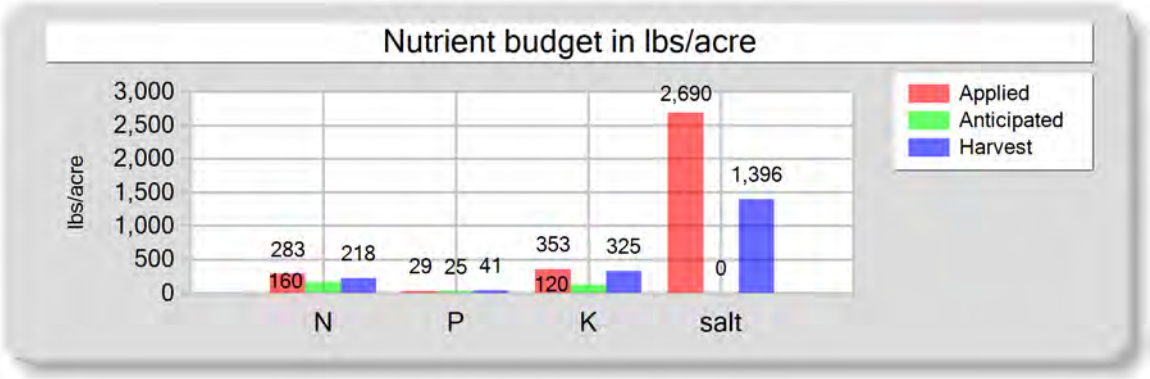
Field 3 - 09/11/2022: Wheat, silage, soft dough

Field name: Field 3 Crop: Wheat, silage, soft dough Plant date: 09/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.55	29.23	352.87	1,987.32
Fresh water	3.16	0.00	0.00	703.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.71	29.23	352.87	2,690.47
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	218.00	41.07	325.43	1,396.42
Nutrient balance	64.71	-11.84	27.44	1,294.05
Applied to removed ratio	1.30	0.71	1.08	1.93

Fresh water applied	
14,382,000.00	*gallons*
529.64	*acre-inches*
13.94	*inches/acre*

Process wastewater applied	
2,220,000.00	*gallons*
81.76	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

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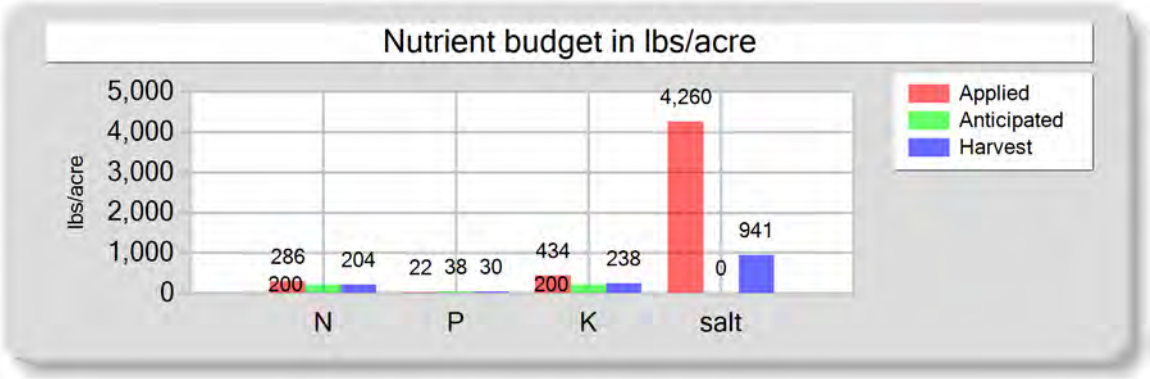
Reporting period 01/01/2023 to 12/31/2023.

Field 3 - 05/26/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.69	22.10	433.79	2,343.51
Fresh water	9.00	0.00	0.00	1,916.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.69	22.10	433.79	4,259.92
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	203.55	29.79	238.31	941.36
Nutrient balance	82.13	-7.68	195.48	3,318.56
Applied to removed ratio	1.40	0.74	1.82	4.53

Fresh water applied	
40,976,760.00	*gallons*
1,509.03	*acre-inches*
39.71	*inches/acre*

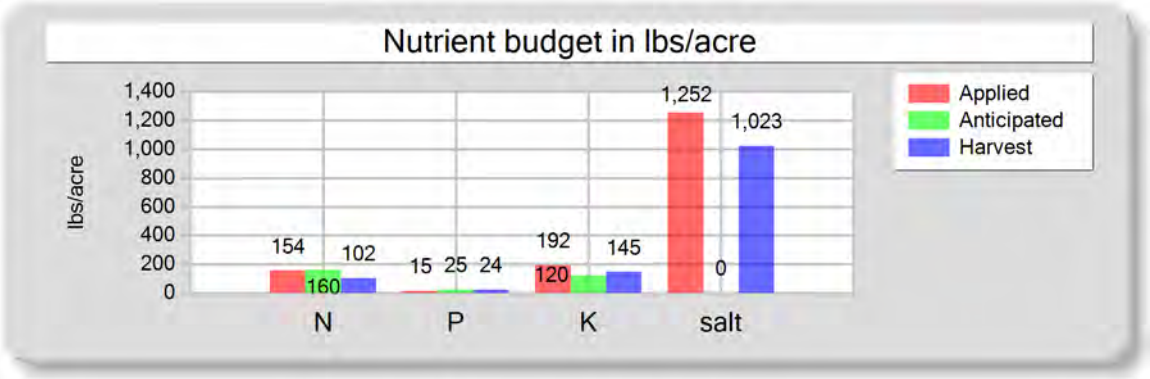
Process wastewater applied	
2,700,000.00	*gallons*
99.43	*acre-inches*
2.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 30 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 30 Crop: Wheat, silage, soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	144.81	15.37	191.71	1,079.54
Fresh water	2.43	0.00	0.00	172.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.24	15.37	191.71	1,252.29
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	102.40	23.81	145.27	1,022.70
Nutrient balance	51.83	-8.45	46.45	229.58
Applied to removed ratio	1.51	0.65	1.32	1.22

Fresh water applied	
12,207,000.00	*gallons*
449.54	*acre-inches*
10.70	*inches/acre*

Process wastewater applied	
1,332,000.00	*gallons*
49.05	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

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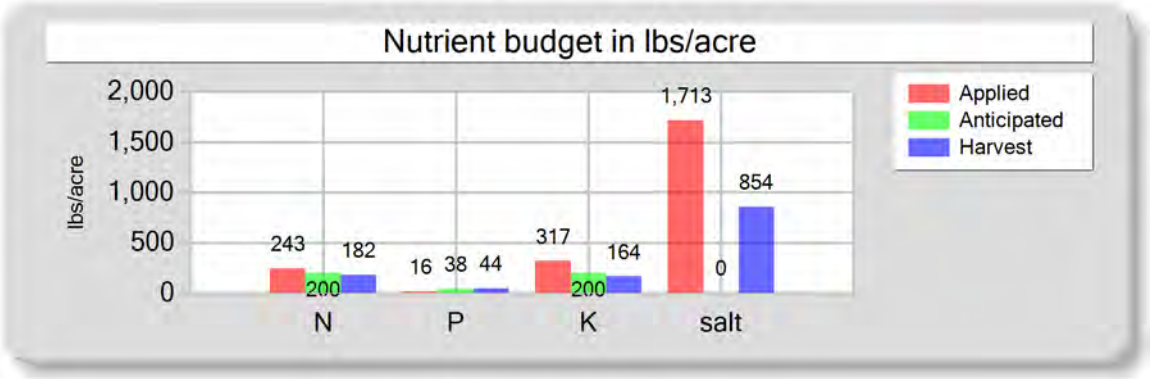
Reporting period 01/01/2023 to 12/31/2023.

Field 30 - 07/01/2023: Corn, silage

Field name: Field 30 Crop: Corn, silage Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	195.63	16.04	316.85	1,713.15
Fresh water	8.81	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.45	16.04	316.85	1,713.15
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	181.88	44.36	164.13	853.93
Nutrient balance	61.57	-28.32	152.72	859.22
Applied to removed ratio	1.34	0.36	1.93	2.01

Fresh water applied	
31,680,000.00	*gallons*
1,166.67	*acre-inches*
38.89	*inches/acre*

Process wastewater applied	
1,548,000.00	*gallons*
57.01	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

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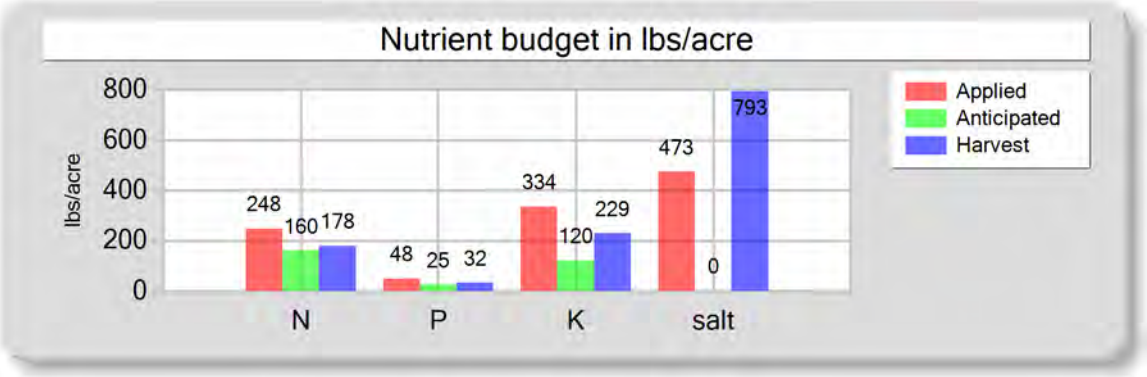
Reporting period 01/01/2023 to 12/31/2023.

Field 31 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 31 Crop: Wheat, silage, soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	158.99	38.95	250.40	0.00
Process wastewater	78.86	9.16	83.88	473.16
Fresh water	2.99	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.84	48.11	334.29	473.16
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	177.96	32.36	228.51	793.14
Nutrient balance	69.88	15.76	105.78	-319.97
Applied to removed ratio	1.39	1.49	1.46	0.60

Fresh water applied	
12,900,000.00	*gallons*
475.06	*acre-inches*
13.20	*inches/acre*

Process wastewater applied	
504,000.00	*gallons*
18.56	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

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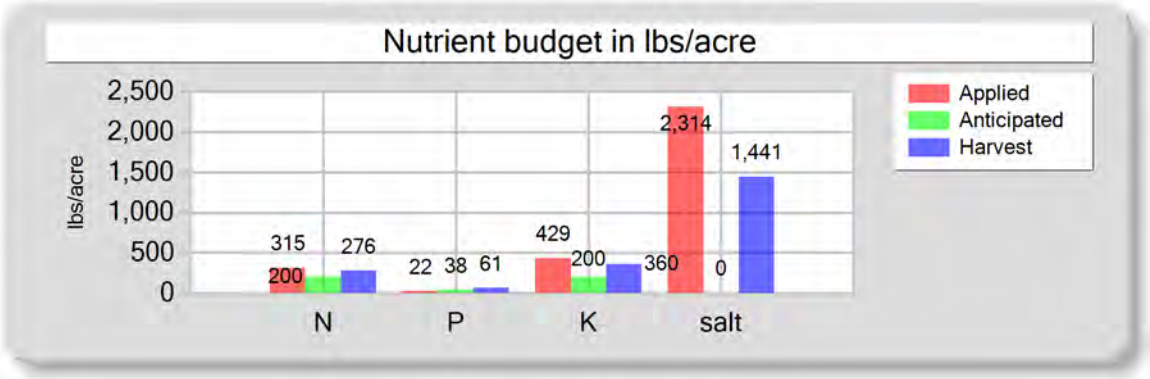
Reporting period 01/01/2023 to 12/31/2023.

Field 31 - 05/15/2023: Corn, silage

Field name: Field 31 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	268.08	21.96	428.63	2,313.96
Fresh water	7.93	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.01	21.96	428.63	2,313.96
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	276.07	60.83	360.30	1,441.47
Nutrient balance	38.94	-38.87	68.33	872.48
Applied to removed ratio	1.14	0.36	1.19	1.61

Fresh water applied	
23,760,000.00	*gallons*
875.00	*acre-inches*
35.00	*inches/acre*

Process wastewater applied	
1,764,000.00	*gallons*
64.96	*acre-inches*
2.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

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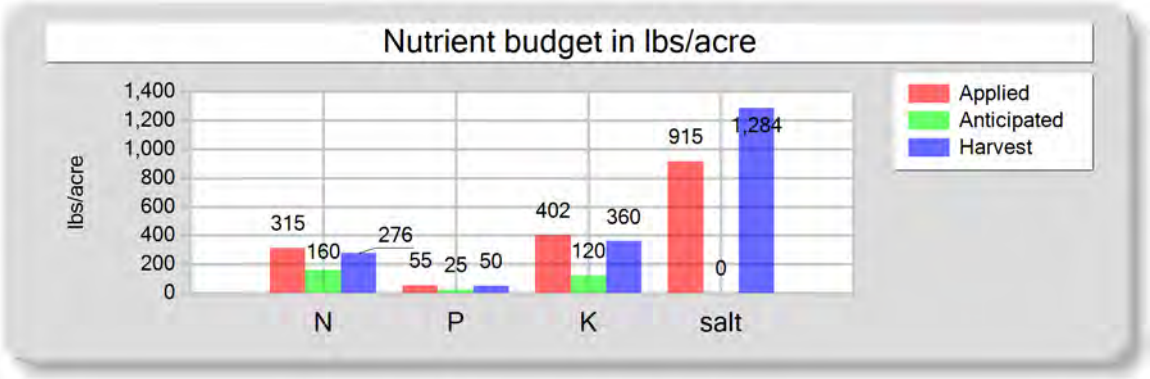
Field 32 - 10/22/2022: Wheat, silage, soft dough

Field name: Field 32 Crop: Wheat, silage, soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.29	37.31	239.85	0.00
Process wastewater	152.50	17.71	162.21	915.00
Fresh water	2.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.54	55.02	402.07	915.00
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	276.25	49.66	360.05	1,283.71
Nutrient balance	38.29	5.36	42.02	-368.71
Applied to removed ratio	1.14	1.11	1.12	0.71

Fresh water applied	
13,500,000.00	*gallons*
497.16	*acre-inches*
12.13	*inches/acre*

Process wastewater applied	
1,110,000.00	*gallons*
40.88	*acre-inches*
1.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

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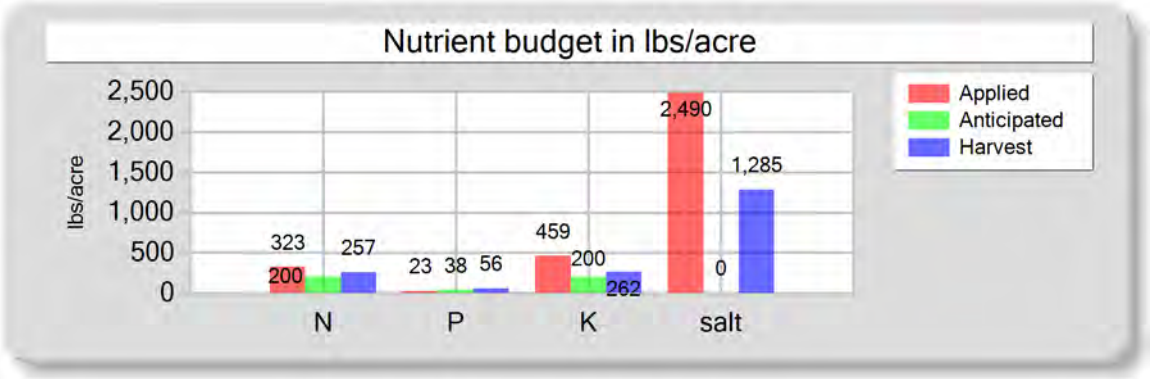
Reporting period 01/01/2023 to 12/31/2023.

Field 32 - 05/15/2023: Corn, silage

Field name: Field 32 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.35	22.72	459.18	2,490.13
Fresh water	7.91	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.26	22.72	459.18	2,490.13
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	257.36	56.15	262.04	1,284.63
Nutrient balance	65.90	-33.43	197.14	1,205.50
Applied to removed ratio	1.26	0.40	1.75	1.94

Fresh water applied	
23,700,000.00	*gallons*
872.79	*acre-inches*
34.91	*inches/acre*

Process wastewater applied	
1,830,000.00	*gallons*
67.39	*acre-inches*
2.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

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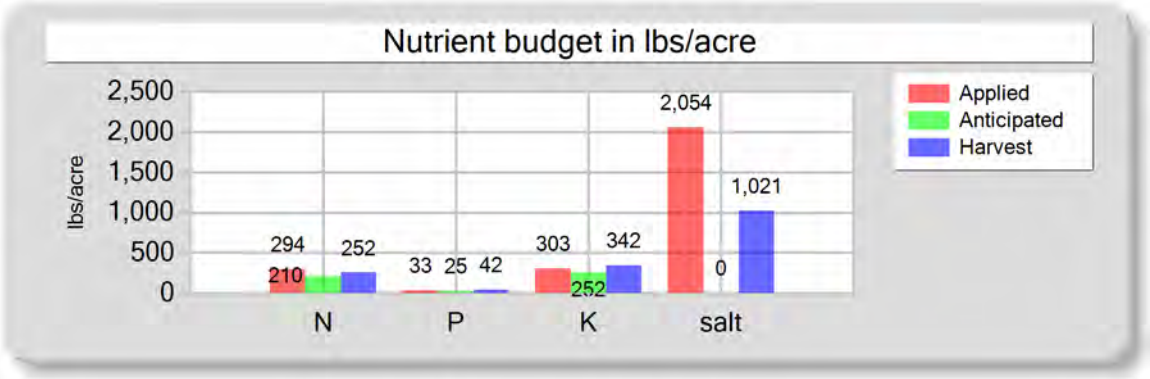
Reporting period 01/01/2023 to 12/31/2023.

Field 4 - 09/11/2022: Rye Silage

Field name: Field 4 Crop: Rye Silage Plant date: 09/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	284.86	33.09	303.01	1,709.18
Fresh water	1.77	0.00	0.00	344.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.63	33.09	303.01	2,053.99
Anticipated crop nutrient removal	210.00	25.20	252.00	0.00
Actual crop nutrient removal	251.72	42.40	341.81	1,020.51
Nutrient balance	41.91	-9.31	-38.80	1,033.48
Applied to removed ratio	1.17	0.78	0.89	2.01

Fresh water applied	
7,416,360.00	*gallons*
273.12	*acre-inches*
7.80	*inches/acre*

Process wastewater applied	
1,770,000.00	*gallons*
65.18	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

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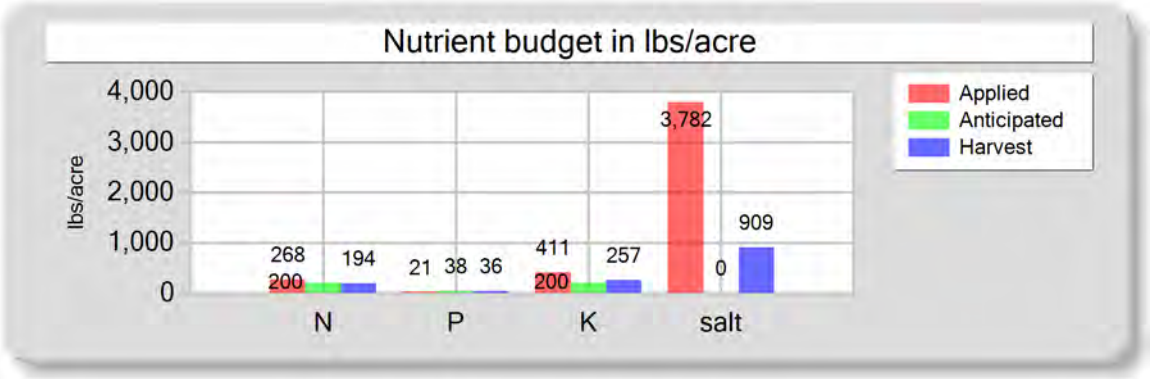
Reporting period 01/01/2023 to 12/31/2023.

Field 4 - 04/24/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 04/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	253.40	20.79	411.20	2,223.82
Fresh water	7.99	0.00	0.00	1,558.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.39	20.79	411.20	3,781.97
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	193.62	36.02	256.66	909.40
Nutrient balance	74.76	-15.24	154.54	2,872.58
Applied to removed ratio	1.39	0.58	1.60	4.16

Fresh water applied	
33,513,240.00	*gallons*
1,234.18	*acre-inches*
35.26	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
2.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

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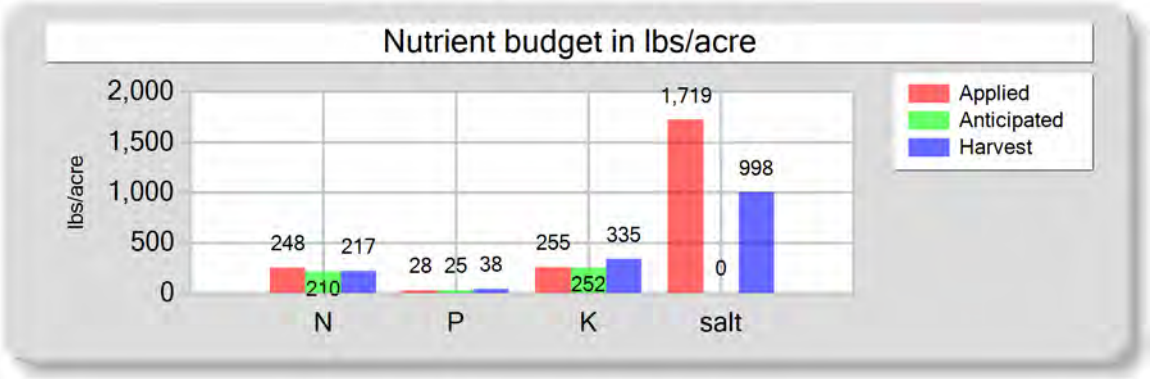
Reporting period 01/01/2023 to 12/31/2023.

Field 5 - 09/11/2022: Rye Silage

Field name: Field 5 Crop: Rye Silage Plant date: 09/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.65	27.84	254.92	1,437.92
Fresh water	1.44	0.00	0.00	281.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.10	27.84	254.92	1,719.43
Anticipated crop nutrient removal	210.00	25.20	252.00	0.00
Actual crop nutrient removal	217.29	37.79	335.38	997.87
Nutrient balance	30.81	-9.95	-80.46	721.56
Applied to removed ratio	1.14	0.74	0.76	1.72

Fresh water applied	
5,708,880.00	*gallons*
210.24	*acre-inches*
6.37	*inches/acre*

Process wastewater applied	
1,404,000.00	*gallons*
51.70	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

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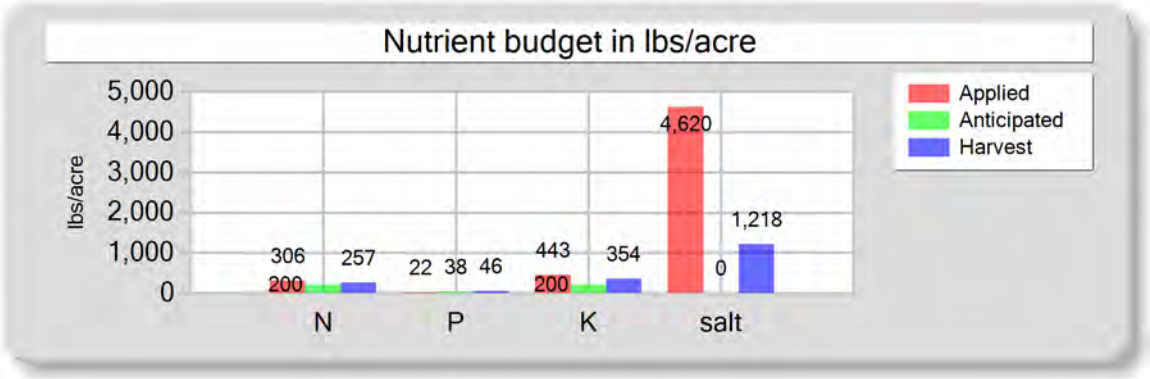
Reporting period 01/01/2023 to 12/31/2023.

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.91	22.39	442.67	2,393.89
Fresh water	25.97	0.00	0.00	2,226.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.87	22.39	442.67	4,620.12
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	257.29	45.74	354.49	1,218.19
Nutrient balance	48.58	-23.35	88.18	3,401.93
Applied to removed ratio	1.19	0.49	1.25	3.79

Fresh water applied	
30,640,860.00	*gallons*
1,128.40	*acre-inches*
34.19	*inches/acre*

Process wastewater applied	
2,376,000.00	*gallons*
87.50	*acre-inches*
2.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

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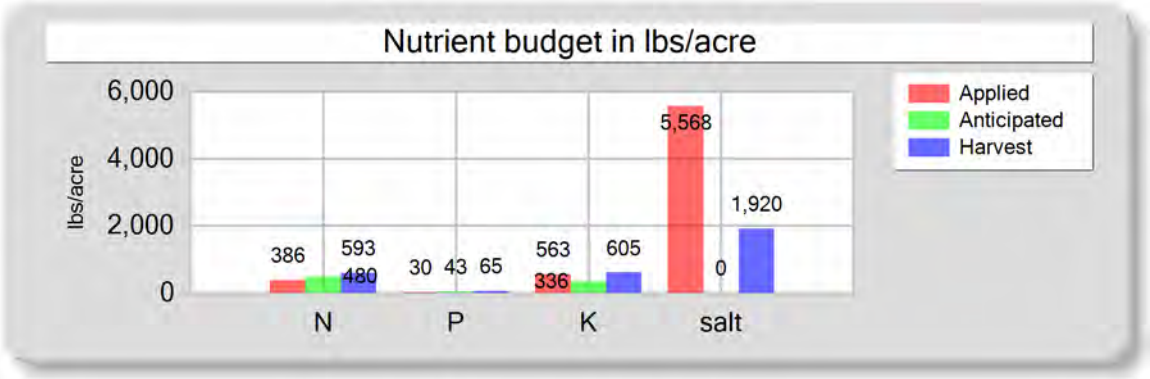
Reporting period 01/01/2023 to 12/31/2023.

Field 6 - 11/01/2021: Alfalfa, hay

Field name: Field 6 Crop: Alfalfa, hay Plant date: 11/01/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	344.02	29.53	562.95	3,081.86
Fresh water	27.83	0.00	0.00	2,486.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	385.85	29.53	562.95	5,567.99
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	593.13	64.60	604.87	1,920.33
Nutrient balance	-207.28	-35.07	-41.92	3,647.66
Applied to removed ratio	0.65	0.46	0.93	2.90

Fresh water applied	
58,679,280.00	*gallons*
2,160.96	*acre-inches*
43.22	*inches/acre*

Process wastewater applied	
4,545,000.00	*gallons*
167.38	*acre-inches*
3.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

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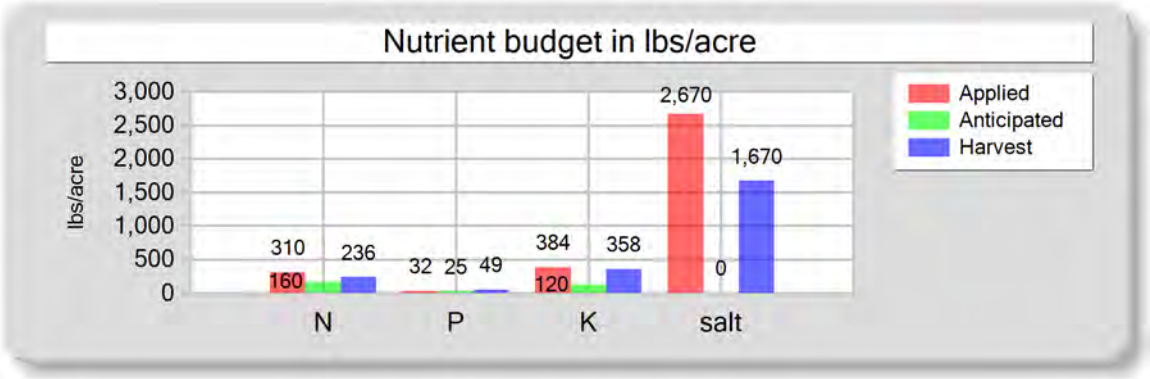
Reporting period 01/01/2023 to 12/31/2023.

Field 7 - 10/15/2022: Wheat, silage, soft dough

Field name: Field 7 Crop: Wheat, silage, soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	300.39	32.41	383.79	2,161.65
Fresh water	2.60	0.00	0.00	507.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	309.99	32.41	383.79	2,669.57
Anticipated crop nutrient removal	159.50	24.65	120.35	0.00
Actual crop nutrient removal	236.20	48.87	358.37	1,669.92
Nutrient balance	73.80	-16.46	25.42	999.65
Applied to removed ratio	1.31	0.66	1.07	1.60

Fresh water applied	
13,109,280.00	*gallons*
482.77	*acre-inches*
11.49	*inches/acre*

Process wastewater applied	
2,670,000.00	*gallons*
98.33	*acre-inches*
2.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

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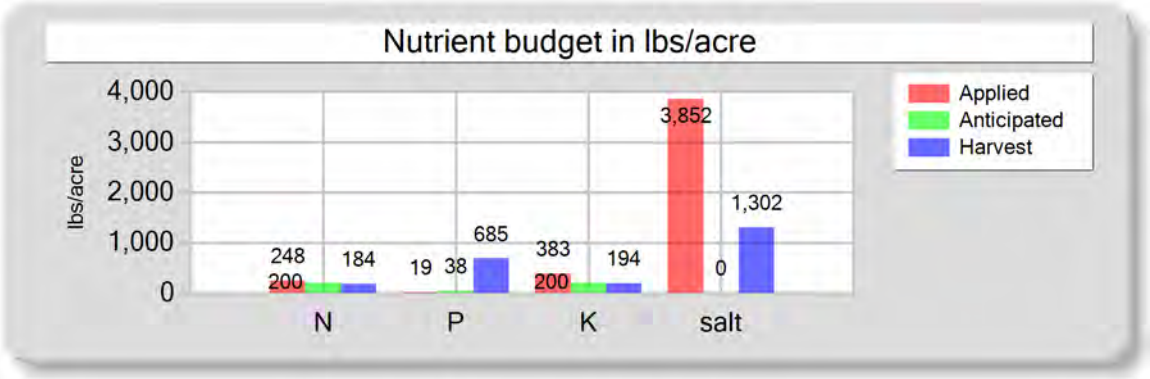
Reporting period 01/01/2023 to 12/31/2023.

Field 7 - 06/08/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.23	19.08	382.68	2,073.32
Fresh water	9.09	0.00	0.00	1,778.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.32	19.08	382.68	3,852.09
Anticipated crop nutrient removal	200.00	37.50	200.00	0.00
Actual crop nutrient removal	184.06	685.12	194.29	1,301.78
Nutrient balance	64.25	-666.05	188.40	2,550.31
Applied to removed ratio	1.35	0.03	1.97	2.96

Fresh water applied	
45,740,400.00	*gallons*
1,684.46	*acre-inches*
40.11	*inches/acre*

Process wastewater applied	
2,580,000.00	*gallons*
95.01	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

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NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids
Sample date: 09/29/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 18.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	4,900.00	31,500.00							0.00
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids
Sample date: 09/29/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	2,000.00	5,600.00							0.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 17.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	3,100.00	20,100.00							0.00
DL	100.00	100.00	100.00							0.01

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Separator Solids

Sample and source description: Separator Solids
Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	3,100.00	20,100.00							0.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids
Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 25.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	4,900.00	40,100.00	20,700.00	9,100.00	6,200.00	4,900.00	0.00		31.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,000.00		0.01

Separator Solids

Sample and source description: Separator Solids
Sample date: 10/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 76.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	4,200.00	33,400.00	19,200.00	7,700.00	5,400.00	4,500.00	0.00		23.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,000.00		0.01

B. PROCESS WASTEWATER ANALYSES

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Slurry

Sample and source description: Slurry

Sample date: 06/02/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3,400.00	1,110.00	0.00	3.70	945.00	1,470.00								11,300.00	3,060
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	455.00	335.00	0.00	3.10	43.40	729.00								7,730.00	4,100
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	100

1QWW

Sample and source description: 1QWW

Sample date: 02/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	675.00	449.00	0.00	1.70	78.40	718.00								6,920.00	4,050
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

2QWW

Sample and source description: 2QWW

Sample date: 06/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	461.00	449.00	0.00	1.30	37.50	681.00								8,040.00	3,640
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

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3QWW

Sample and source description: 3QWW

Sample date: 09/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,400.00	413.00	0.00	0.50	333.00	795.00	566.00	334.00	262.00	4,720.00	0.00	35.90	264.00	8,140.00	2,220
DL	1.00	0.50	0.50	0.10	0.10	0.50	0.10	0.10	1.00	10.00	1.00	0.50	0.20	10.00	10

3QWW

Sample and source description: 3QWW

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	447.00	402.00	0.00	1.40	37.00	796.00								7,770.00	4,350
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	671.00	418.00	0.00	0.70	47.00	1,040.00								9,820.00	5,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Irrigation Wells

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Irrigation Wells

IW #1

Sample description: IW #1

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								308.00	200
DL	1.00	0.50	0.10								10.00	10

IW #10

Sample description: IW #10

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								404.00	243
DL	1.00	0.50	0.10								10.00	10

IW #11

Sample description: IW #11

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								402.00	243
DL	1.00	0.50	0.10								10.00	10

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Irrigation Wells

IW #12

Sample description: IW #12

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.00	48
DL	1.00	0.50	0.10								10.00	10

IW #2

Sample description: IW #2

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	27.80	0.00	27.80								1,530.00	1,000
DL	1.00	0.50	0.10								10.00	10

IW #3

Sample description: IW #3

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								551.00	355
DL	1.00	0.50	0.10								10.00	10

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Irrigation Wells

IW #4

Sample description: IW #4

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								308.00	195
DL	1.00	0.50	0.10								10.00	10

IW #5

Sample description: IW #5

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								598.00	335
DL	0.10	0.50	0.10								10.00	10

IW #6

Sample description: IW #6

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								595.00	345
DL	0.10	0.50	0.10								10.00	10

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Irrigation Wells

IW #8

Sample description: IW #8

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								276.00	182
DL	1.00	0.50	0.10								10.00	10

IW #9

Sample description: IW #9

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								276.00	185
DL	1.00	0.50	0.10								10.00	10

IW Borba

Sample description: IW Borba

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.00								326.00	0
DL	1.00	0.50	0.10								10.00	10

MID Canal

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MID Canal

MID Canal

Sample description: MID Canal

Sample date: 08/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								402.00	248
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/15/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,500.00	1,300.00	11,000.00		8.40
DL	100.00	100.00	100.00		0.01

Field 1 - 05/26/2023: Corn, silage

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Field 1 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	900.00	5,600.00		5.60
DL	100.00	100.00	100.00		0.01

Field 10-11 - 10/26/2022: Wheat Hay

Winter Forage Hay

Sample and source description: Winter Forage Hay
Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 11.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,000.00	9,900.00		13.50
DL	100.00	100.00	100.00		0.01

Field 10-11 - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	900.00	3,500.00		4.60
DL	100.00	100.00	100.00		0.01

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Field 12 - 10/01/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa
Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	2,900.00	15,300.00		8.40
DL	100.00	100.00	100.00		0.01

Field 15 - 10/20/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	900.00	4,500.00		10.60
DL	100.00	100.00	100.00		0.01

Field 15 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	900.00	3,200.00		6.00
DL	100.00	100.00	100.00		0.01

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Field 16 - 10/20/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,000.00	5,300.00		9.60
DL	100.00	100.00	100.00		0.01

Field 16 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	900.00	2,900.00		5.80
DL	100.00	100.00	100.00		0.01

Field 17 - 09/25/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	2,900.00	15,300.00		9.50
DL	100.00	100.00	100.00		0.01

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Field 18 - 10/12/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa
Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,900.00	3,300.00	23,700.00		16.05
DL	100.00	100.00	100.00		0.01

Field 19 - 10/21/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 51.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	1,500.00	8,100.00		8.50
DL	100.00	100.00	100.00		0.01

Field 19 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	800.00	2,800.00		4.70
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 2 - 10/15/2022: Rye Silage

Rye Silage

Sample and source description: Rye Silage
Sample date: 04/11/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	1,800.00	14,900.00		13.65
DL	100.00	100.00	100.00		0.01

Field 2 - 05/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,000.00	8,800.00		7.90
DL	100.00	100.00	100.00		0.01

Field 20 - 10/21/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,100.00	5,600.00		10.60
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 20 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	700.00	2,900.00		4.90
DL	100.00	100.00	100.00		0.01

Field 21 - 10/21/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	1,200.00	5,100.00		8.60
DL	100.00	100.00	100.00		0.01

Field 21 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	900.00	3,100.00		5.40
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 29 - 10/22/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	1,000.00	5,900.00		10.30
DL	100.00	100.00	100.00		0.01

Field 29 - 07/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	1,000.00	3,500.00		4.60
DL	100.00	100.00	100.00		0.01

Field 3 - 09/11/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 54.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,300.00	10,300.00		9.80
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 3 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	600.00	4,800.00		6.70
DL	100.00	100.00	100.00		0.01

Field 30 - 10/22/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 59.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	1,000.00	6,100.00		10.50
DL	100.00	100.00	100.00		0.01

Field 30 - 07/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	3,700.00		5.00
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 31 - 10/22/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	1,600.00	11,300.00		9.10
DL	100.00	100.00	100.00		0.01

Field 31 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 57.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,300.00	7,700.00		7.30
DL	100.00	100.00	100.00		0.01

Field 32 - 10/22/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,900.00	1,600.00	11,600.00		12.20
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 32 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	5,600.00		7.40
DL	100.00	100.00	100.00		0.01

Field 4 - 09/11/2022: Rye Silage

Rye Silage

Sample and source description: Rye Silage

Sample date: 04/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	1,600.00	12,900.00		13.10
DL	100.00	100.00	100.00		0.01

Field 4 - 04/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	800.00	5,700.00		6.60
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 5 - 09/11/2022: Rye Silage

Rye Silage

Sample and source description: Rye Silage
Sample date: 04/11/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	1,600.00	14,200.00		12.50
DL	100.00	100.00	100.00		0.01

Field 5 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	800.00	6,200.00		6.70
DL	100.00	100.00	100.00		0.01

Field 6 - 11/01/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa
Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,300.00	3,300.00	30,900.00		10.90
DL	100.00	100.00	100.00		0.01

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

Field 7 - 10/15/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	1,200.00	8,800.00		10.10
DL	100.00	100.00	100.00		0.01

Field 7 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	13,400.00	3,800.00		6.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

No subsurface (tile) drainage analyses entered.

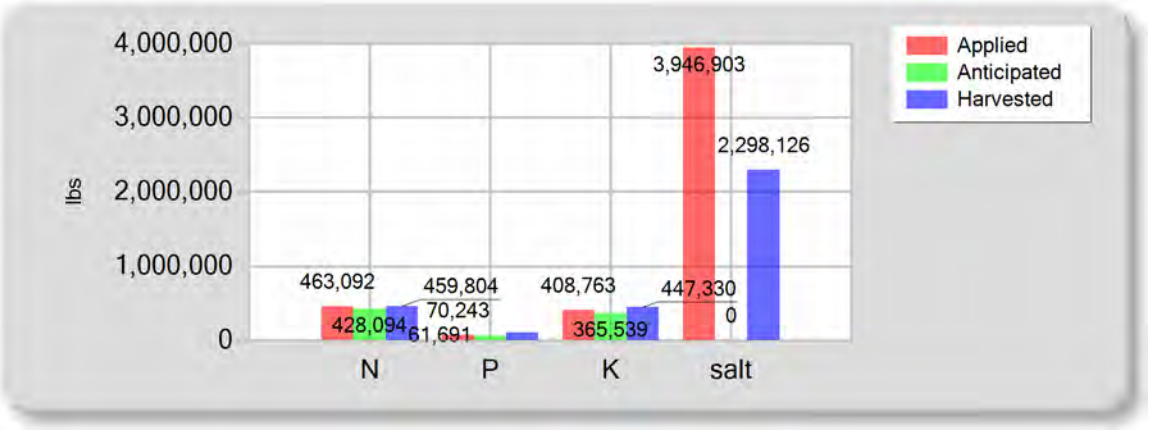
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

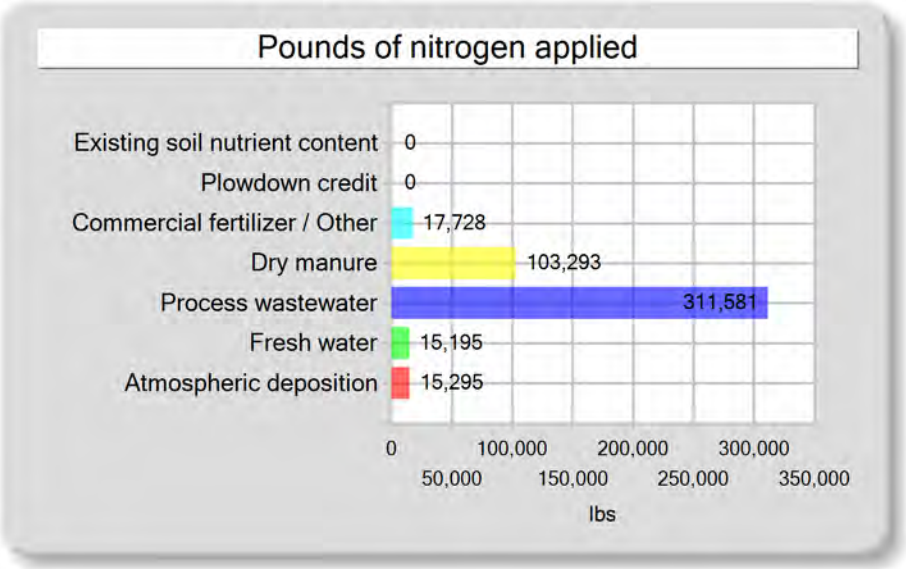
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	17,728.00	0.00	0.00	0.00
Dry manure	103,292.72	16,400.42	94,499.69	0.00
Process wastewater	311,581.37	53,842.36	314,263.13	1,534,561.30
Fresh water	15,195.35	0.00	0.00	2,412,342.05
Atmospheric deposition	15,295.00	0.00	0.00	0.00
Total nutrients applied	463,092.44	70,242.78	408,762.82	3,946,903.35
Anticipated crop nutrient removal	428,093.50	61,690.75	365,538.55	0.00
Actual crop nutrient removal	459,803.70	104,598.89	447,329.93	2,298,125.62
Nutrient balance	3,288.74	-34,356.11	-38,567.11	1,648,777.74
Applied to removed ratio	1.01	0.67	0.91	1.72

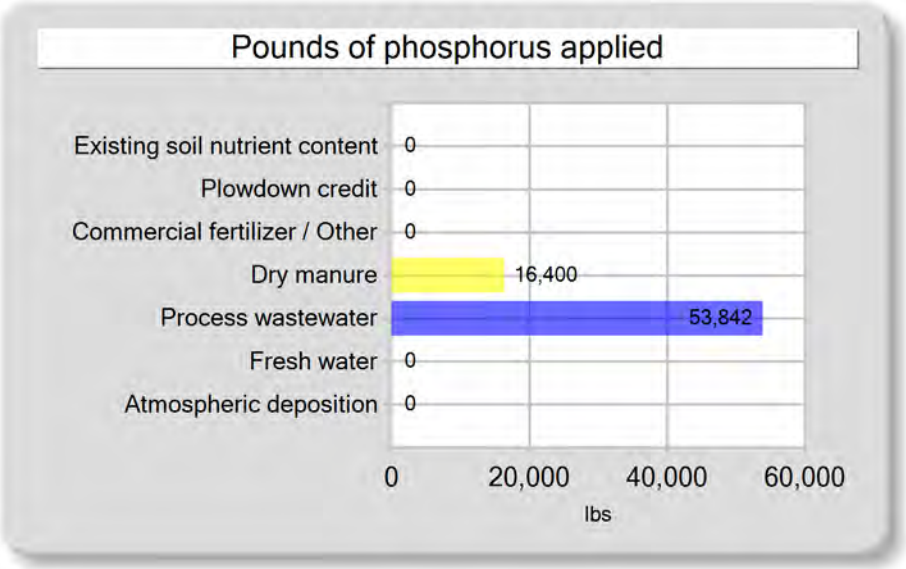
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

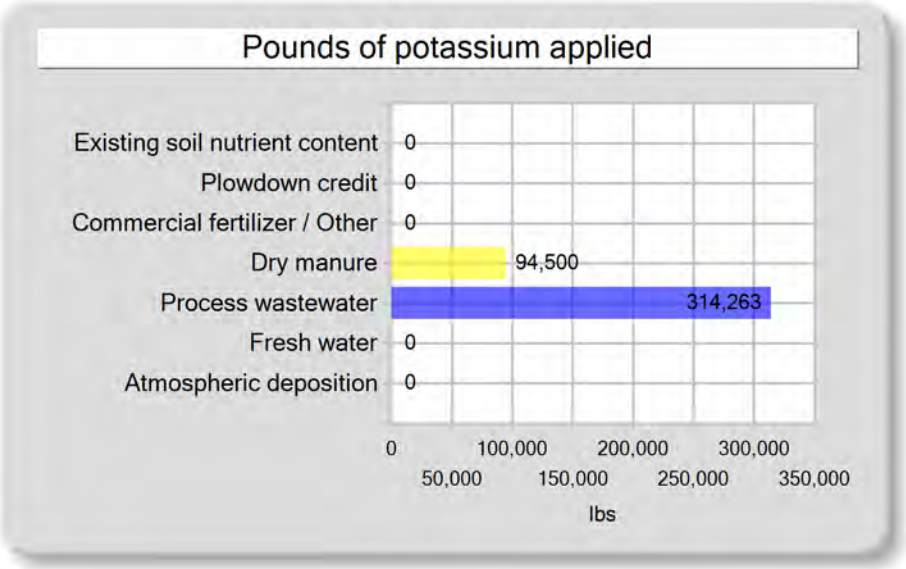


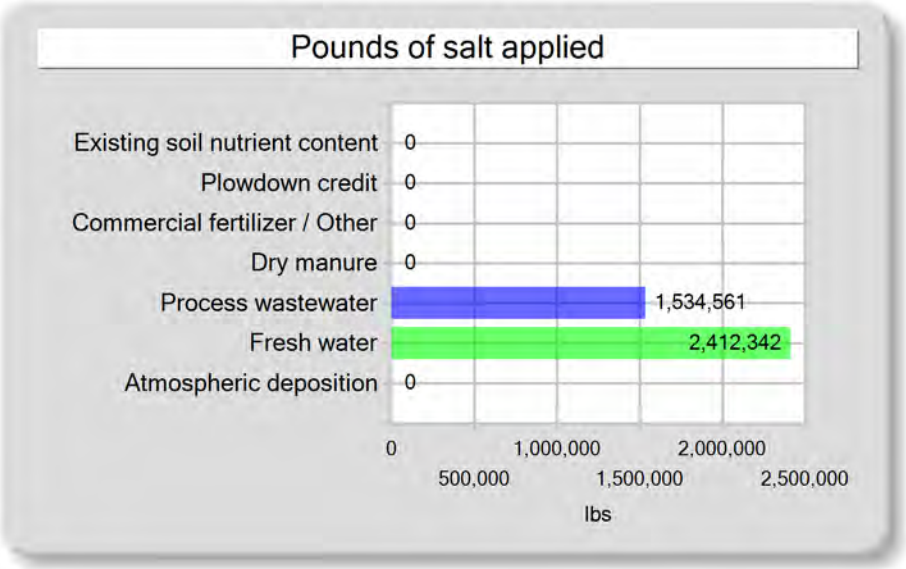
Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ADDITIONAL NOTES

A. NOTES

Precipitation utilized during winter months to meet forage freshwater requirements.

Irrigation well #7 is non operational for 2023 and will be sampled once it becomes operational. Heavy rains during the winter season allowed for a greater amount of surface water allocation to grow crops.

Field 1 Wheat had a higher than anticipated removal rate. This was due to a higher than expected %N and yield. The %N removed, which is based on analysis derived through a certified laboratory, exceeded the expected values for this crop. Appropriate management practices resulted in decent yields and higher than expected nutrient removal rates.

Fields 10/11 Wheat Hay, 15 Wheat, 20 Wheat, 21 Wheat, & 29 Wheat had lower than anticipated removal rates. This was due to lower than expected yields and/or a lower than expected %N. The %N was based on analysis that was derived through a certified laboratory. However, the applications to these fields matched the low removal rates and was able to meet the field ratio threshold of 1.4.

Fields 29 Corn & 30 Wheat Silage had lower than anticipated removal rates due to lower than anticipated %N &/or extremely low tonnage. This resulted in field ratios slightly exceeding target limits.

Field 12 Alfalfa received no nutrients in 2023. All nutrients applied to this field was contributed through freshwater applications only.

Field 8 was fallow during the 2023 cropping season.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

DocuSigned by:
59609E2E7E0947E...

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Case Vyfhuisen	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/20/2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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DELLAVALLE™
LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:36
Reported: 08/28/2023 10:52

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1826-01	5H 1	Ground Water	Case V.		08/21/2023 0:00
23H1826-02	5H 2	Ground Water	Case V.		08/21/2023 0:00
23H1826-03	5H 3	Ground Water	Case V.		08/21/2023 0:00
23H1826-04	5H 4	Ground Water	Case V.		08/21/2023 0:00
23H1826-05	5H 5	Ground Water	Case V.		08/21/2023 0:00
23H1826-06	5H 6	Ground Water	Case V.		08/21/2023 0:00
23H1826-07	5H 8	Ground Water	Case V.		08/21/2023 0:00
23H1826-08	5H 9	Ground Water	Case V.		08/21/2023 0:00
23H1826-09	5H 10	Ground Water	Case V.		08/21/2023 0:00
23H1826-10	5H 11	Ground Water	Case V.		08/21/2023 0:00

Default Cooler
Containers Intact
COC/Labels Agree
Received On Ice

Temperature on Receipt °C: 6.3

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:36
Reported: 08/28/2023 10:52

Sample Results

Sample: 5H 1
23H1826-01 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.31**	mmhos/cm	0.01	1		08/23/23 12:22	SM 2510 B		BEH1082
Electrical Conductivity umhos	**308**	umhos/cm	10.0	1		08/23/23 12:22	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 19:16	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:22	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**200**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:22	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 12:58	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 12:58	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:36
Reported: 08/28/2023 10:52

Sample Results
(Continued)

Sample: 5H 2
23H1826-02 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.53**	mmhos/cm	0.01	1		08/23/23 12:23	SM 2510 B		BEH1082
Electrical Conductivity umhos	**1530**	umhos/cm	10.0	1		08/23/23 12:23	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	**27.8**	mg/L	0.1	1	10	08/21/23 19:36	EPA 300.0		BEH0987
pH	**7.9**	units	1.0	1		08/23/23 12:23	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**1000**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:23	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 12:59	SM 4500-NH3 C		BEH1135
Total Nitrogen	**27.8**	mg/L	1.00	1		08/25/23 12:59	SM 4500-NH3 C		BEH1135

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Account Manager: Ben Nydam
Submitted By: FR Ag Services

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Sample Results
(Continued)

Sample: 5H 3
23H1826-03 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		08/23/23 12:24	SM 2510 B		BEH1082
Electrical Conductivity umhos	**551**	umhos/cm	10.0	1		08/23/23 12:24	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 19:56	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:24	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**355**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:24	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:01	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:01	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 4
23H1826-04 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.31**	mmhos/cm	0.01	1		08/23/23 12:26	SM 2510 B		BEH1082
Electrical Conductivity umhos	**308**	umhos/cm	10.0	1		08/23/23 12:26	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 20:17	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:26	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**195**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:26	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:02	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:02	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 5
23H1826-05 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.60**	mmhos/cm	0.01	1		08/23/23 12:27	SM 2510 B		BEH1082
Electrical Conductivity umhos	**598**	umhos/cm	10.0	1		08/23/23 12:27	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	08/21/23 20:37	EPA 300.0		BEH0987
pH	**8.3**	units	1.0	1		08/23/23 12:27	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**335**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:27	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:04	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:04	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 6
23H1826-06 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.60**	mmhos/cm	0.01	1		08/23/23 12:28	SM 2510 B		BEH1082
Electrical Conductivity umhos	**595**	umhos/cm	10.0	1		08/23/23 12:28	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	08/21/23 20:57	EPA 300.0		BEH0987
pH	**8.3**	units	1.0	1		08/23/23 12:28	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**345**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:28	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:05	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:05	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 8
23H1826-07 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		08/23/23 12:36	SM 2510 B		BEH1082
Electrical Conductivity umhos	**276**	umhos/cm	10.0	1		08/23/23 12:36	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 21:17	EPA 300.0		BEH0987
pH	**8.1**	units	1.0	1		08/23/23 12:36	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**182**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:36	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:06	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:06	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 9
23H1826-08 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		08/23/23 12:37	SM 2510 B		BEH1082
Electrical Conductivity umhos	**276**	umhos/cm	10.0	1		08/23/23 12:37	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 21:38	EPA 300.0		BEH0987
pH	**8.1**	units	1.0	1		08/23/23 12:37	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**185**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:37	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:08	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:08	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 10
23H1826-09 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.40**	mmhos/cm	0.01	1		08/23/23 12:38	SM 2510 B		BEH1082
Electrical Conductivity umhos	**404**	umhos/cm	10.0	1		08/23/23 12:38	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 21:58	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:38	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**243**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:38	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:09	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:09	SM 4500-NH3 C		BEH1135

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Sample Results
(Continued)

Sample: 5H 11
23H1826-10 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.40**	mmhos/cm	0.01	1		08/23/23 12:39	SM 2510 B		BEH1082
Electrical Conductivity umhos	**402**	umhos/cm	10.0	1		08/23/23 12:39	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/21/23 22:19	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:39	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**243**	mg/L	10.0	1		08/24/23 12:07	SM 2540 C		BEH1056
Temperature	**25.0**	°C	0.0	1		08/23/23 12:39	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:19	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:19	SM 4500-NH3 C		BEH1135

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0987										
Blank (BEH0987-BLK1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0987-BLK2)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0987-BLK3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0987-BS1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEH0987-BS2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEH0987-DUP1)		**Source: 23H1826-01**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			0.00	10
Duplicate (BEH0987-DUP2)		**Source: 23H1828-02**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			3.39	10
Matrix Spike (BEH0987-MS1)		**Source: 23H1826-01**			Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.03	104	90-110		
Matrix Spike (BEH0987-MS2)		**Source: 23H1828-02**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.03	104	90-110		
Reference (BEH0987-SRM1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0987-SRM2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEH0987-SRM3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1056										
Blank (BEH1056-BLK1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1056-BS1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	31.2		10.0	mg/L	2000		1.56	0-200		
Duplicate (BEH1056-DUP1)			**Source: 23H1762-01**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	200		10.0	mg/L		198			1.26	10
Duplicate (BEH1056-DUP2)			**Source: 23H1829-01**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	810		10.0	mg/L		800			1.24	10
Reference (BEH1056-SRM1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	320			mg/L	325.0		98.5	90-110		
Reference (BEH1056-SRM2)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	487			mg/L	495.0		98.3	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082										
Blank (BEH1082-BLK1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1082-BLK2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1082-BLK3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	6.1		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1082-DUP1)		**Source: 23H1823-14**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.0		1.0	units		7.9			0.126	10
Electrical Conductivity	0.79		0.01	mmhos/cm		0.80			1.28	10
Electrical Conductivity umhos	792		10.0	umhos/cm		802			1.28	10
Duplicate (BEH1082-DUP2)		**Source: 23H1829-01**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.0		1.0	units		8.0			0.00	10
Electrical Conductivity	1.17		0.01	mmhos/cm		1.16			0.249	10
Electrical Conductivity umhos	1170		10.0	umhos/cm		1160			0.249	10
Reference (BEH1082-SRM1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	522			umhos/cm	538.0		97.0	90-110		
Reference (BEH1082-SRM2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEH1082-SRM3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM4)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	983			umhos/cm	1000		98.3	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082 (Continued)										
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity umhos	983			umhos/cm	1000		98.3	90-110		
Reference (BEH1082-SRM6)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM7)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM8)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:36
Reported: 08/28/2023 10:52

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1135										
Blank (BEH1135-BLK1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1135-BLK2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1135-BS1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.92		1.00	mg/L	5.709		104	90-110		
LCS (BEH1135-BS2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1135-DUP1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH1135-DUP2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1135-MS1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	8.30		1.40	mg/L	7.992	ND	104	90-110		
Matrix Spike (BEH1135-MS2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	7.73		1.40	mg/L	7.992	ND	96.7	90-110		
Reference (BEH1135-SRM1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	23.3			mg/L	23.80		97.9	90-110		

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WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.
1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: 22207 Acct # | 8 Cons #

Results Need By

No. Samples: 10 No of Bottles: ~~10~~ 20

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [X] Groundwater [ ] Monitoring Well

Other:

Name: 5H Farms
Address: 5851 S Highway 59
City: Merced **State:** CA **Zip:** 95341
Telephone: **Fax:**
Cell/Email: c5huizen07@gmail.com
COPY TO: labs@fragservices.com
krhodes@fragservices; ariordan@fragservices.com
REQUESTED BY: FR Ag Services
PROJECT:
CROP:

[X] Copy of Chain [X] QA/QC Documents
Sampled By: ~~FR Ag Services~~ Case V.

Analysis and Bottles Required: (*Please indicate Analysis*)

( ) DWW1: EC, NO_3-N — NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	5H 1	8/21		6.3	
2	5H 2			7.1	
3	5H 3			6.5	
4	5H 4			5.8	
5	5H 5			6.4	
6	5H 6			5.6	
7	5H 8			6.1	
8	5H 9			6.1	
9	5H 10			5.8	
10	5H 11			5.8	

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Kierstan Rhodes	FR Ag Services	8/21 0800	8/21 1000
Second	JLF	DLV	8-21-23 12:38	8-21-23 2:00
Third	MM	DLI	8/21/23 14:36	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payme Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $ If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Labo Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to bin bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor reasonable attorneys' fees of Dellavalle Laboratory.

08/21/23 14:36 23H1826

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[5H Farms Water .xls]Dairy Water

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08/21/23 14:36 23H1826

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☐ DLI Sampler ☑ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic	1	1	1	1	1	1	1	1	1	1
Plastics	* pH Value	<2	<2	<2	<2	<2	<2	<2	<2	<2	<2
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic	1	1	1	1	1	1	1	1	1	1
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SEND OUT

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DELLAVALLE™
LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1828-01	5H 12	Drinking Water	Case V.		08/21/2023 0:00
23H1828-02	5H D New	Drinking Water	Case V.		08/21/2023 0:00
23H1828-03	5H Borba	Drinking Water	Case V.		08/21/2023 0:00
23H1828-04	Mid Water	Drinking Water	Case V.		08/21/2023 0:00

Default Cooler Temperature on Receipt °C: 5.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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5H Farms
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Merced, CA 95341

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Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Sample Results

Sample: 5H 12
23H1828-01 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.40**	mmhos/cm	0.01	1		08/23/23 12:41	SM 2510 B		BEH1082
Electrical Conductivity umhos	**402**	umhos/cm	10.0	1		08/23/23 12:41	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/22/23 01:02	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:41	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**248**	mg/L	10.0	1		08/24/23 09:12	SM 2540 C		BEH1058
Temperature	**25.0**	°C	0.0	1		08/23/23 12:41	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:20	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:20	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Sample Results
(Continued)

Sample: 5H D New
23H1828-02 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.32**	mmhos/cm	0.01	1		08/23/23 12:42	SM 2510 B		BEH1082
Electrical Conductivity umhos	**316**	umhos/cm	10.0	1		08/23/23 12:42	SM 2510 B		BEH1082
Ammonia (as N)	*	mg/L	0.00	1		08/21/23 00:00	Field		BEH1300
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/22/23 01:22	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:42	SM 4500-H+	H	BEH1082
Temperature	**25.0**	°C	0.0	1		08/23/23 12:42	SM 2510 B		BEH1082

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Sample Results
(Continued)

Sample: 5H Borba
23H1828-03 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.33**	mmhos/cm	0.01	1		08/23/23 12:43	SM 2510 B		BEH1082
Electrical Conductivity umhos	**326**	umhos/cm	10.0	1		08/23/23 12:43	SM 2510 B		BEH1082
Ammonia (as N)	*	mg/L	0.00	1		08/21/23 00:00	Field		BEH1300
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/22/23 01:42	EPA 300.0		BEH0987
pH	**8.2**	units	1.0	1		08/23/23 12:43	SM 4500-H+	H	BEH1082
Temperature	**25.0**	°C	0.0	1		08/23/23 12:43	SM 2510 B		BEH1082

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

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Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Sample Results
(Continued)

Sample: Mid Water
23H1828-04 (Water)

Sampled: 8/21/2023 0:00
Sampled By: Case V.

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.03**	mmhos/cm	0.01	1		08/23/23 12:45	SM 2510 B		BEH1082
Electrical Conductivity umhos	**29.6**	umhos/cm	10.0	1		08/23/23 12:45	SM 2510 B		BEH1082
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	08/22/23 02:03	EPA 300.0		BEH0987
pH	**7.3**	units	1.0	1		08/23/23 12:45	SM 4500-H+	H	BEH1082
Total Filterable Solids (TDS)	**48.0**	mg/L	10.0	1		08/24/23 09:12	SM 2540 C		BEH1058
Temperature	**25.0**	°C	0.0	1		08/23/23 12:45	SM 2510 B		BEH1082
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/25/23 13:21	SM 4500-NH3 C		BEH1135
Total Nitrogen	ND	mg/L	1.00	1		08/25/23 13:21	SM 4500-NH3 C		BEH1135

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

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Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0987										
Blank (BEH0987-BLK1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0987-BLK2)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0987-BLK3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0987-BS1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEH0987-BS2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEH0987-DUP1)		**Source: 23H1826-01**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			0.00	10
Duplicate (BEH0987-DUP2)		**Source: 23H1828-02**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	0.03		0.1	mg/L		0.03			3.39	10
Matrix Spike (BEH0987-MS1)		**Source: 23H1826-01**			Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.03	104	90-110		
Matrix Spike (BEH0987-MS2)		**Source: 23H1828-02**			Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.03	104	90-110		
Reference (BEH0987-SRM1)					Prepared & Analyzed: 8/21/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH0987-SRM2)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEH0987-SRM3)					Prepared & Analyzed: 8/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1058										
Blank (BEH1058-BLK1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1058-BS1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	26.2		10.0	mg/L	2000		1.31	0-200		
Duplicate (BEH1058-DUP1)			**Source: 23H1834-02**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	720		10.0	mg/L		710			1.40	10
Duplicate (BEH1058-DUP2)			**Source: 23H1845-01**		Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	140		10.0	mg/L		150			6.90	10
Reference (BEH1058-SRM1)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	327			mg/L	325.0		101	90-110		
Reference (BEH1058-SRM2)					Prepared: 8/22/2023	Analyzed: 8/24/2023				
Total Filterable Solids (TDS)	490			mg/L	495.0		99.0	90-110		

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Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082										
Blank (BEH1082-BLK1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1082-BLK2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1082-BLK3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	6.1		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEH1082-DUP1)		**Source: 23H1823-14**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.0		1.0	units		7.9			0.126	10
Electrical Conductivity	0.79		0.01	mmhos/cm		0.80			1.28	10
Electrical Conductivity umhos	792		10.0	umhos/cm		802			1.28	10
Duplicate (BEH1082-DUP2)		**Source: 23H1829-01**			Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	8.0		1.0	units		8.0			0.00	10
Electrical Conductivity	1.17		0.01	mmhos/cm		1.16			0.249	10
Electrical Conductivity umhos	1170		10.0	umhos/cm		1160			0.249	10
Reference (BEH1082-SRM1)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	522			umhos/cm	538.0		97.0	90-110		
Reference (BEH1082-SRM2)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	5.8			units	5.820		100	28178-101.71		
Reference (BEH1082-SRM3)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM4)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	980			umhos/cm	1000		98.0	90-110		
Electrical Conductivity umhos	980			umhos/cm	1000		98.0	90-110		
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity	983			umhos/cm	1000		98.3	90-110		

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DELLAVALLE™ LABORATORY INC

5H Farms
5851 S Highway 59
Merced, CA 95341

Account# 00-0022207
Account Manager: Ben Nydam
Submitted By: FR Ag Services

Received: 08/21/2023 14:32
Reported: 08/28/2023 13:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1082 (Continued)										
Reference (BEH1082-SRM5)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
Electrical Conductivity umhos	983			umhos/cm	1000		98.3	90-110		
Reference (BEH1082-SRM6)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM7)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH1082-SRM8)					Prepared: 8/22/2023	Analyzed: 8/23/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1135										
Blank (BEH1135-BLK1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1135-BLK2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1135-BS1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.92		1.00	mg/L	5.709		104	90-110		
LCS (BEH1135-BS2)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1135-DUP1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH1135-DUP2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1135-MS1)		**Source: 23H1826-09**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	8.30		1.40	mg/L	7.992	ND	104	90-110		
Matrix Spike (BEH1135-MS2)		**Source: 23H1829-03**			Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	7.73		1.40	mg/L	7.992	ND	96.7	90-110		
Reference (BEH1135-SRM1)					Prepared: 8/23/2023	Analyzed: 8/25/2023				
Kjeldahl Nitrogen (TKN), Total	23.3			mg/L	23.80		97.9	90-110		

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WATER WORK REQUEST

DELLAVALLE LABORATORY, INC
1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

8-21-23

Purchase Order No

Bill To: 22207 Acct # | 8 Cons #

No. Samples: 4 No of Bottles: 6

Results Need By

Water Type: [X] Drinking Water [ ] Wastewater [X] Ag Water [X] Groundwater [ ] Monitoring Well

Other:

Name: 5H Farms
Address: 5851 S Highway 59
City: Merced **State:** CA **Zip:** 95341
Telephone: **Fax:**
Cell/Email: c5huizen07@gmail.com
COPY TO: labs@fragservices.com
krhodes@fragservices; ariordan@fragservices.com
REQUESTED BY: FR Ag Services
PROJECT:
CROP:

[X] Copy of Chain [X] QA/QC Documents
Sampled By: FR Ag Services Case V.

Analysis and Bottles Required: (*Please indicate Analysis*)

(X) DWW1: EC, NO_3-N NH4-N Field Test
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TKN, TN, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples		Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	SH 12	Canal DCW1	8/21		5.8	
2	SH D New	DWW1			7.1	
3	SH Borba	DWW1			7.0	
4	MID water	Canal DCW1			6.3	
5						
6						
7						
8						
9						
10						

08/21/23 14:32 23H1828

EDS

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Kierstan Rhodes	FR Ag Services	8/21 0800	8/21 1000
Second	JLF	DLV	8-21-23 12:38	8-21-23 2:00
Third	MM	DLV	8/21/23 14:32	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Shipping

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/
Location: Laborato

X:\1-CLIENT CUSTOM FIELDSHEETS\FIELDSHEETS\Dairy Fieldsheets\[5H Farms Water .xls]Dairy Water

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08/21/23 14:32 23H1828

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☐ DLI Sampler ☑ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value	<2			<2						
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1						
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

SENT OUT